

AI+RES Event Slate

Nine CONUS cases run through rare-event resampling at week-3 lead, ranging from a heat dome that no direct ensemble reaches to a day sitting at the ensemble median. Each case is reported with its observed anomaly, its percentile against 64 years of ERA5, and the probability the run assigns to it.

Every number on this page reduces to one observable:

A_L = $\cos(\text{lat})$ -weighted mean, over the event box, of the 7-day-mean 2 m temperature anomaly ending at the event peak

GenCast is initialised 21 days before that peak and rolled as 64 walkers. At each of five barriers FourCastNet 3 scores every walker's expected A_L . Walkers with high scores are cloned, the rest are killed, and a weight w_i corrects for the resulting tilt, so the surviving population still estimates GenCast's own probabilities. The C schedule (0, 1.0, 1.4, 1.8, 2.0) was tuned once, on the Pacific Northwest case, and has not been re-tuned since.

The severity ladder

Cases are ordered by how far the observation fell outside GenCast's own week-3 forecast distribution. That distance, rather than the size of the anomaly or its climatological rarity, predicts whether resampling reaches the observation. Winter Storm Elliott at -14.42 K is reached. The Southwest heat wave at $+5.14$ K is not, because it lies 0.5σ deeper in the model's spread.

RUNG	EVENT	REGIME	OBSERVED A_L	CLIM. PCT	RETURN PERIOD	Σ - DEPTH	DIRECT	AI+RES	WEIGHTED P	OUTCOME
1	Southwest Heat Wave August 2020	heat	+5.14 K	99.96th	210 yr	+3.3 σ	0 / 24	0 / 64	$< 3 \times 10^{-4}$	not reached
2	California Heat Wave September 2022	heat	+4.71 K	99.71th	37 yr	$\sim +2.9 \sigma$	0 / 24	25 / 64	0.0186	reached
3	Winter Storm Elliott December 2022	cold	-14.42 K	98.41th	5 yr	+2.81 σ	0 / 24	20 / 64	0.0079	reached
4	Winter Storm Uri February 2021	cold	-7.40 K	96.80th	4 yr	+2.28 σ	0 / 24	32 / 64	0.0090	reached
5	South-Central Heat Dome June 2023	heat	+5.73 K	100.00th	beyond the record	$\sim +2.2 \sigma$	0 / 16	6 / 64	0.00273	reached

RUNG	EVENT	REGIME	OBSERVED A _L	CLIM. PCT	RETURN PERIOD	Σ- DEPTH	DIRECT	AI+RES	WEIGHTED P	OUTCOME
6	Pacific Northwest Heat Dome June 2021	heat	+7.72 K	99.60th	22 yr	+1.98 σ	0 / 40	42 / 64	0.054	reached
7	The Nationwide Warm Spell 24 December 2025	CONUS index	+3.26 K	98.78th	10 yr	+1.91 σ	1 / 24	46 / 64	0.0661	anchor
8	The Normal-Day Test 2 August 2024	CONUS index	+1.06 K	94.08th	4 yr	0.00 σ	14 / 24	63 / 64	0.649	control
9	The Below-Median Day 7 November 2023	CONUS index	+0.21 K	60.10th	1 yr	-0.97 σ	21 / 24	64 / 64	0.583 (pinned)	pinned

The two percentile columns answer different questions. The climatological percentile measures how rare the weather was, from the same index computed over 64 years of ERA5 and pooled within ±21 days of the event's calendar position. The σ-depth measures how hard the event was to forecast, from where the observation fell in the 24-member GenCast ensemble launched three weeks earlier. The South-Central heat dome is the rarest case in the table and only the fifth hardest to reach.

The whole population

all ten runs
CROSS-EVENT

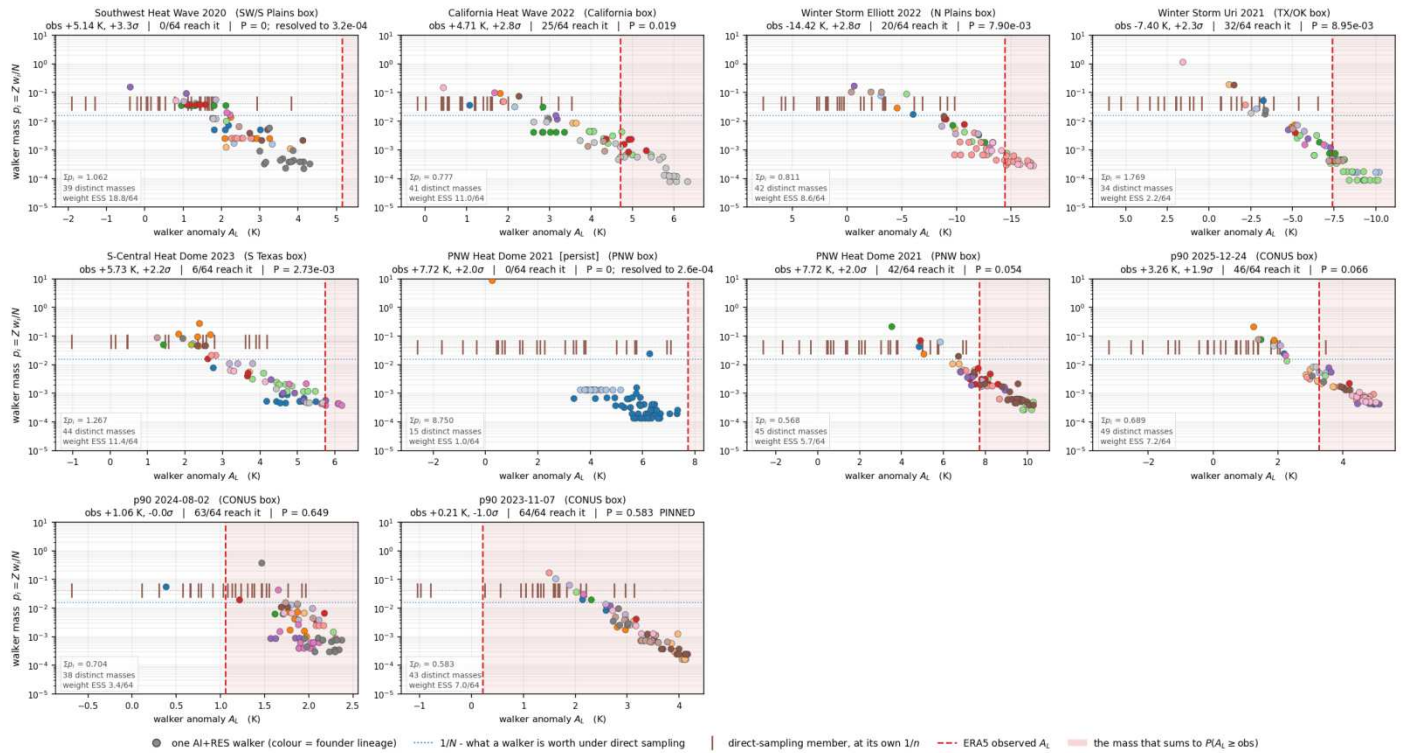
Every number in the severity-ladder table is a sum over the walkers of one run. The population figure below gives that raw material, one panel per run, in the same hardest-first order. Each panel shows where the 64 surviving walkers landed on A_L and the probability mass $p_i = Z \cdot w_i / N$ that each carries. The shaded band is the mass that sums to the headline P .

Deeper reach costs probability mass. The blue line is $1/N$, the mass a walker would carry had it not been resampled, and the deepest walkers lie one to two decades below it. The hottest Pacific Northwest walker carries 2.5 % of the mass of a single direct member.

The masses come in tied blocks. Walkers cloned at the last barrier inherit their parent's mass exactly, so the 64 walkers of Winter Storm Uri hold 34 distinct values and those of the persistence control hold 15. That count, not 64, is the number of independent pieces of information behind the estimate.

Two runs cannot answer their own question. The Southwest heat wave and the persistence control have no walker in the band, so P is 0 and the statement the run supports is the deepest level it resolved, 3.2 and 2.6×10^{-4} respectively. The 7 November 2023 case fails in the opposite way. Its observation lies below every walker, so the band covers the whole panel and P equals the run's own normalization check.

AI+RES: every surviving walker, its anomaly, and the probability it carries
each panel's headline P is the mass right of the red line; $\sum p_i$ over the whole panel is the run's normalization check



Panels ordered by how far the observation sat outside the DIRECT ensemble's spread, hardest first. A walker far below the 1/N line is one resampling manufactured and the weights are charging for; walkers stacked at one height are clones of a single parent at the last resampling and carry identical mass by construction. Two runs cannot answer their own question: Southwest and the persistence control reached nothing, so P is 0 and the honest statement is the deepest level the run resolved, and p90 2023-11-07 is PINNED - the observation lies below every walker, so the subset sum is the whole panel and P equals the normalization check, not a probability.

The population, all ten runs. One panel per run, ordered by how far the observation fell outside the direct ensemble's spread. Each dot is a walker, coloured by founder lineage. The brown ticks are the direct-sampling members at their own 1/n. The red dashed line is the observation, and the shaded side of it is the mass that sums to P . $\sum p_i$ over a whole panel is that run's normalization check, which should equal 1 and is quoted in each panel's corner alongside the weight ESS.

The trajectory panel in each case section also carries the operational CFSv2 forecast at the same 21-day lead, for all nine events. CFSv2 reaches the observation on the two mildest rungs only, with 3 of 4 members on 7 November 2023 and 2 of 4 on 2 August 2024, and misses every extreme case. It misses the 24 December 2025 warm spell by 4.6 K, with its ensemble mean on the opposite side of climatology from the observation. That miss is not a resolution artefact. Regridding the ERA5 truth field onto the CFS $\sim 0.94^\circ$ grid and re-scoring it moves A_L by at most 0.13 K, against gaps of 2.3 to 11.6 K.

Southwest Heat Wave

August 2020

RUNG 1 OF 9

heat not reached

The deepest target relative to GenCast's own spread, and the one case the run does not reach. The observation is $+3.3 \sigma$ out, and a tilt with $C \leq 2$ and $K = 5$ moves the population mean by about $+1.75 \text{ K}$, not by the $+4.4 \text{ K}$ required.

OBSERVED A_L

+5.14 K ERA5, 29-35 N, 109-103 W

CLIMATOLOGICAL PERCENTILE

99.96th ERA5 1959-2022 · 100.00th against 1990-2019

RETURN PERIOD

210 yr 1 of 64 years (the event year itself) holds a window this extreme

DEPTH IN GENCAST'S SPREAD

+3.3 σ week-3 ensemble $+0.71 \pm 1.33 \text{ K}$

DIRECT SAMPLING

0 / 24 best member $+3.82 \text{ K}$ · $P = 0.000$ [0.000, 0.142]

AI+RES

0 / 64 most extreme $+4.31 \text{ K}$

WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$

$< 3 \times 10^{-4}$ sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

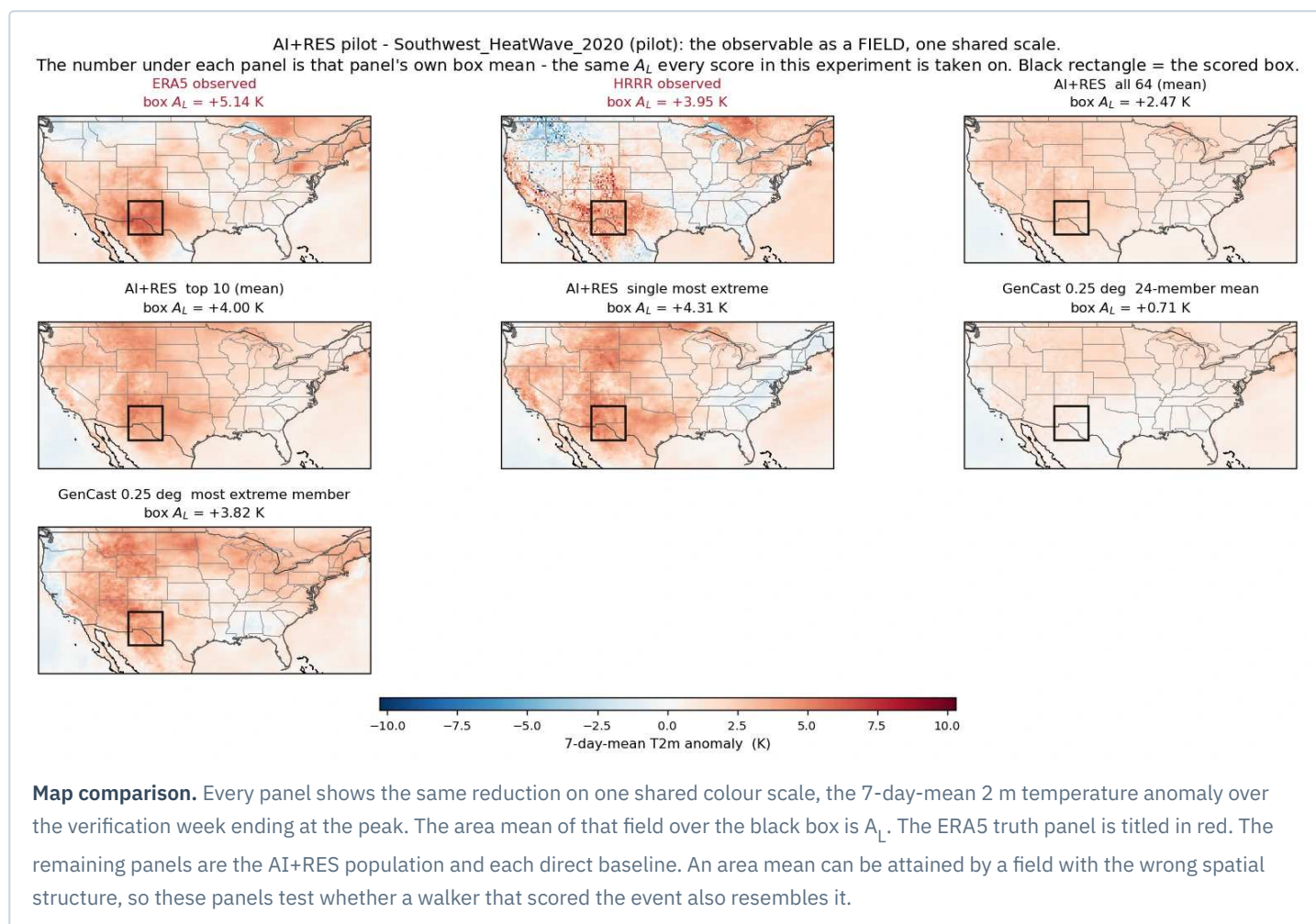
+4.16 K the single day, not the week · 99.1th percentile

64-YEAR RECORD FOR THIS SEASON

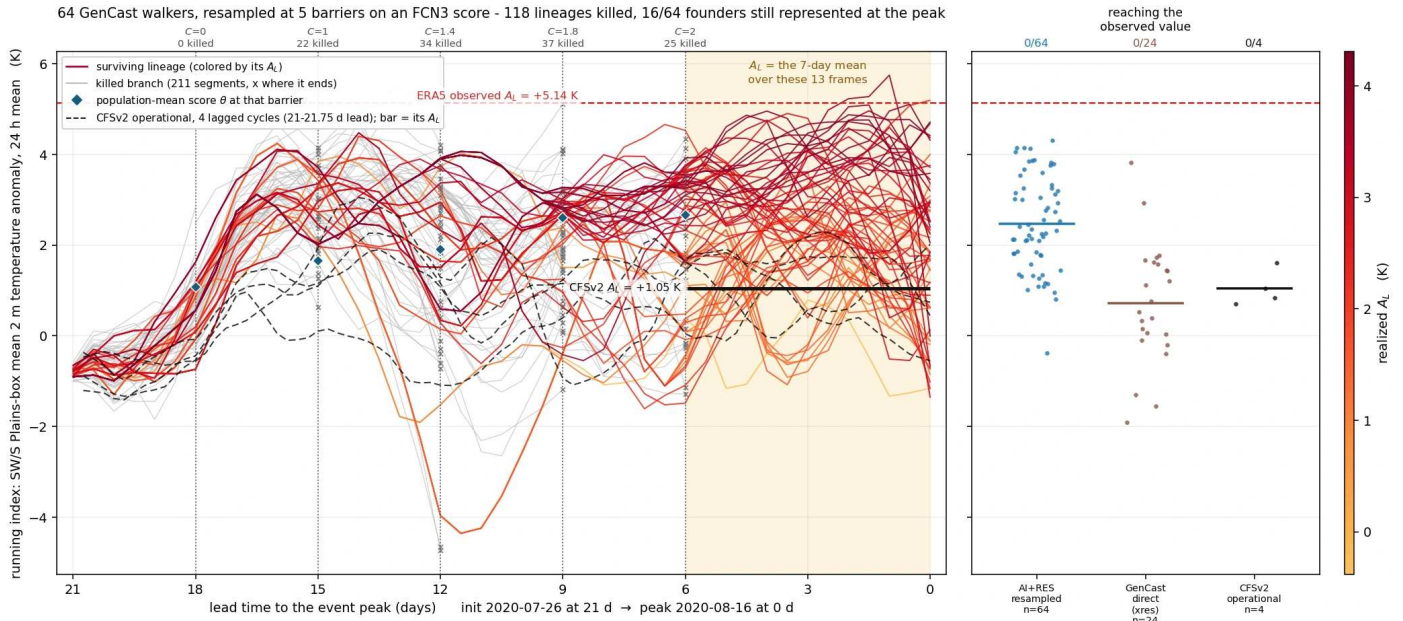
+5.15 K - this event peak 2020-08-16 · init 2020-07-26 (21 d lead)

box: Southern New Mexico and west Texas (29-35 N, 109-103 W) · registry key Southwest_HeatWave_2020 · Slurm job 1176, 7 h 34 m
run health: log Z -0.8660 · normalization check 1.0623 · 16/64 founders · ESS/N crashed to 0.11 at step 3 (max multiplicity 23), recovering to 0.25 and 0.37

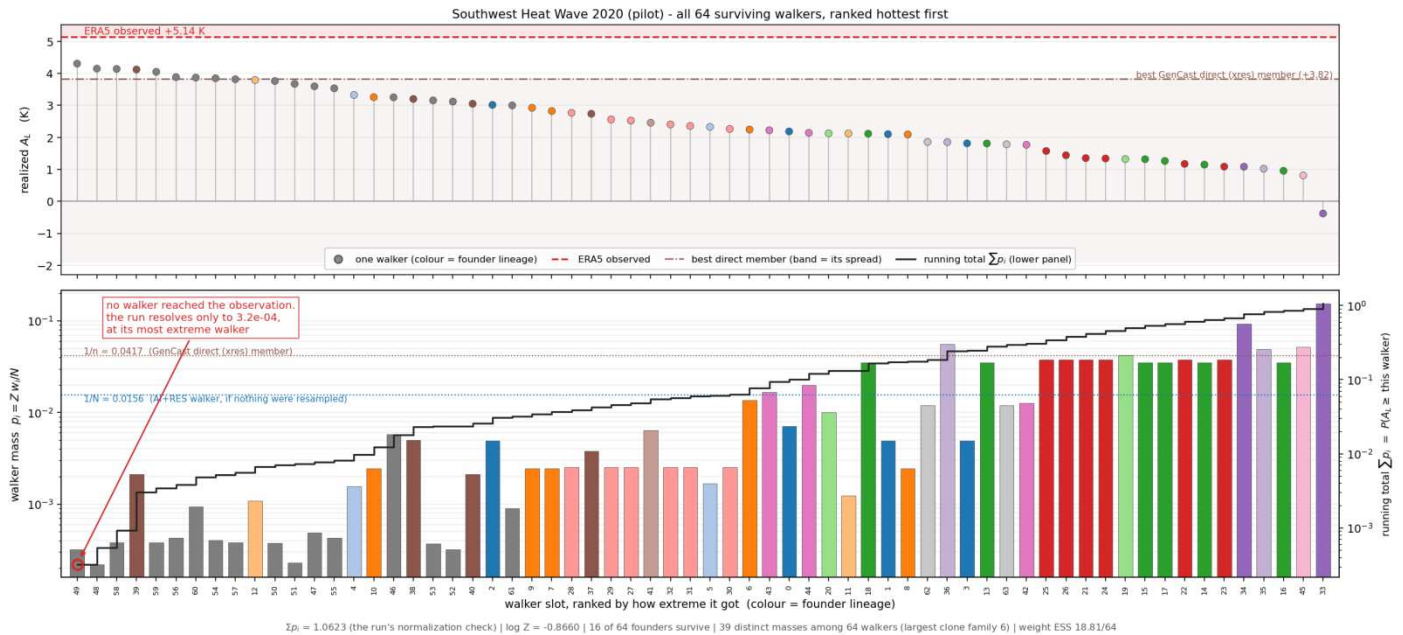
The exceedance curve stays calibrated over the range the 24 direct members resolve. The weighted exceedance agrees with direct sampling to within binomial noise, giving 0.077 against 0.083 at $A_L \geq +2.23$ K and 0.026 against 0.042 at $A_L \geq +3.03$ K, and it then extends about two decades deeper. The weights remove most of the tilt, reducing the population mean from +2.47 K unweighted to +1.24 K weighted.



AI+RES pilot - Southwest_HeatWave_2020 (pilot): the walk itself, barrier by barrier



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.

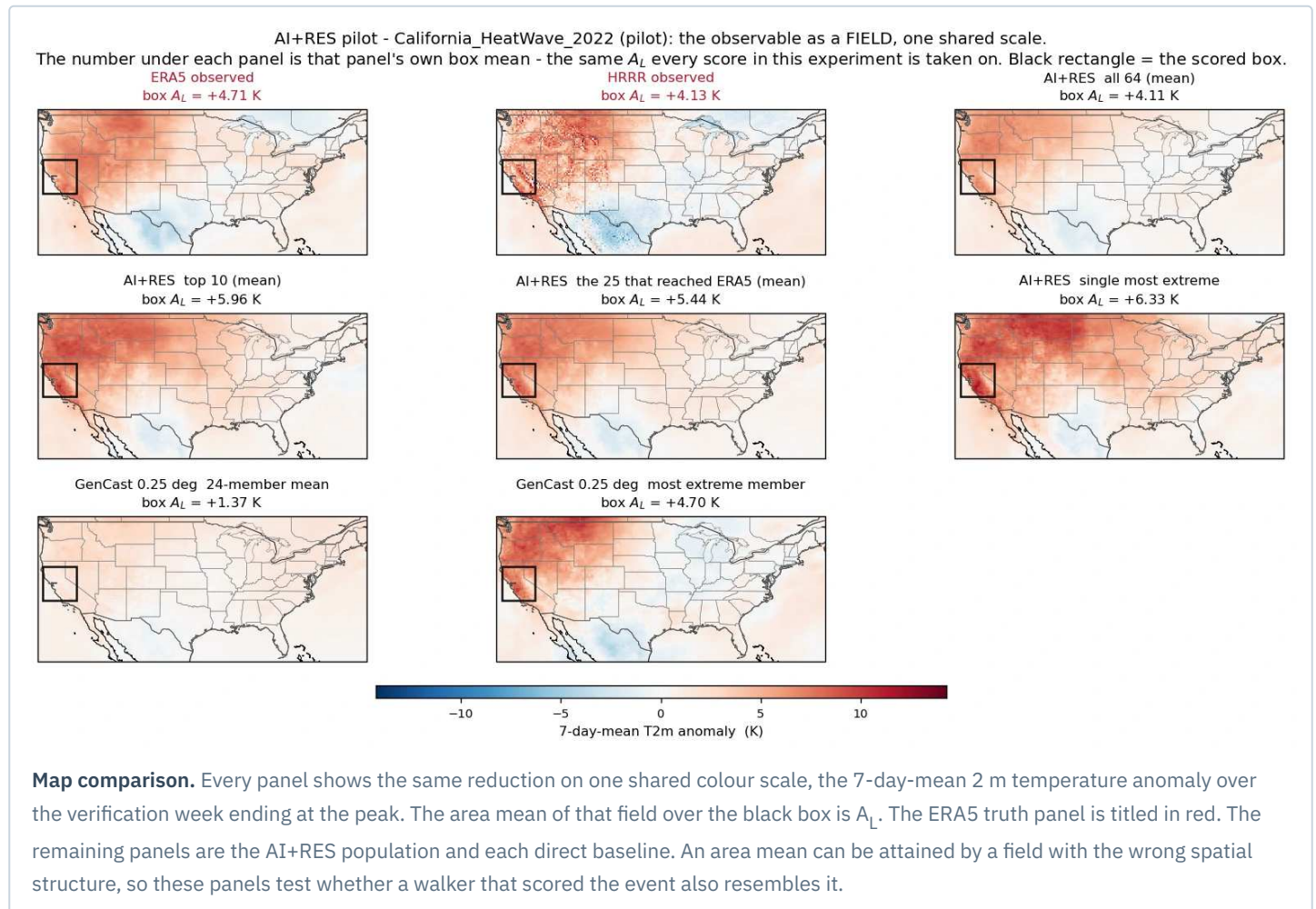


Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

A reach at moderate severity. The best direct-sampling member reached +4.70 K without crossing the observation. 25 of 64 walkers crossed it.

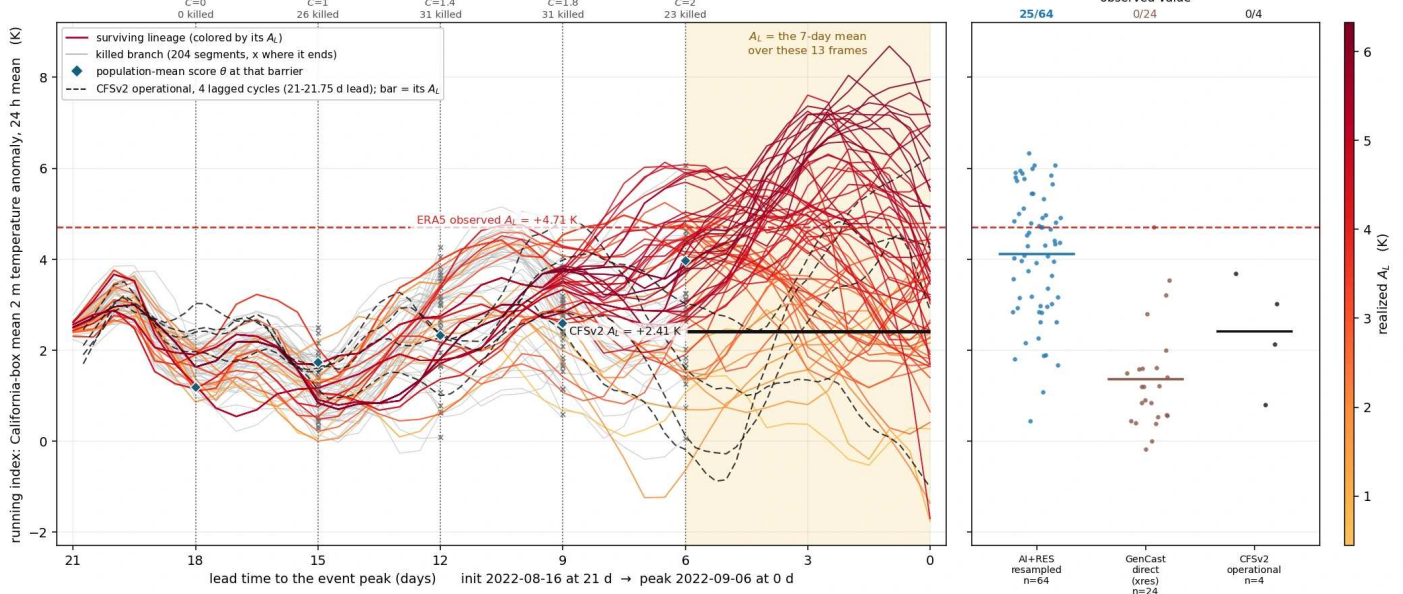
OBSERVED A_L +4.71 KERA5, 35-41 N, 124-118 W
CLIMATOLOGICAL PERCENTILE 99.71thERA5 1959-2022 · 99.84th against 1990-2019
RETURN PERIOD 37 yr2 of 64 (2017, 2022) years hold a window this extreme
DEPTH IN GENCAST'S SPREAD ~+2.9 σweek-3 ensemble not tabulated
DIRECT SAMPLING 0 / 24best member +4.70 K · P = 0.000 [0.000, 0.142]
AI+RES 25 / 64most extreme +6.33 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.0186sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +7.42 Kthe single day, not the week · 100.0th percentile
64-YEAR RECORD FOR THIS SEASON +6.30 K (2022-09-10)peak 2022-09-06 · init 2022-08-16 (21 d lead)

Sacramento's all-time record of 116 °F fell on this peak. The box covers the named region rather than the location of the week's maximum. The unconstrained maximum for that week is +7.32 K over Idaho and Montana, and a box placed there would score a different event from the one the case names. Where direct sampling resolves the curve the weighted estimate agrees with it, giving 0.114 against 0.125 at $A_L \geq +2.99$ K and 0.045 against 0.042 at $A_L \geq +3.99$ K. The curve extends to 7.7×10^{-5} at +6.33 K.



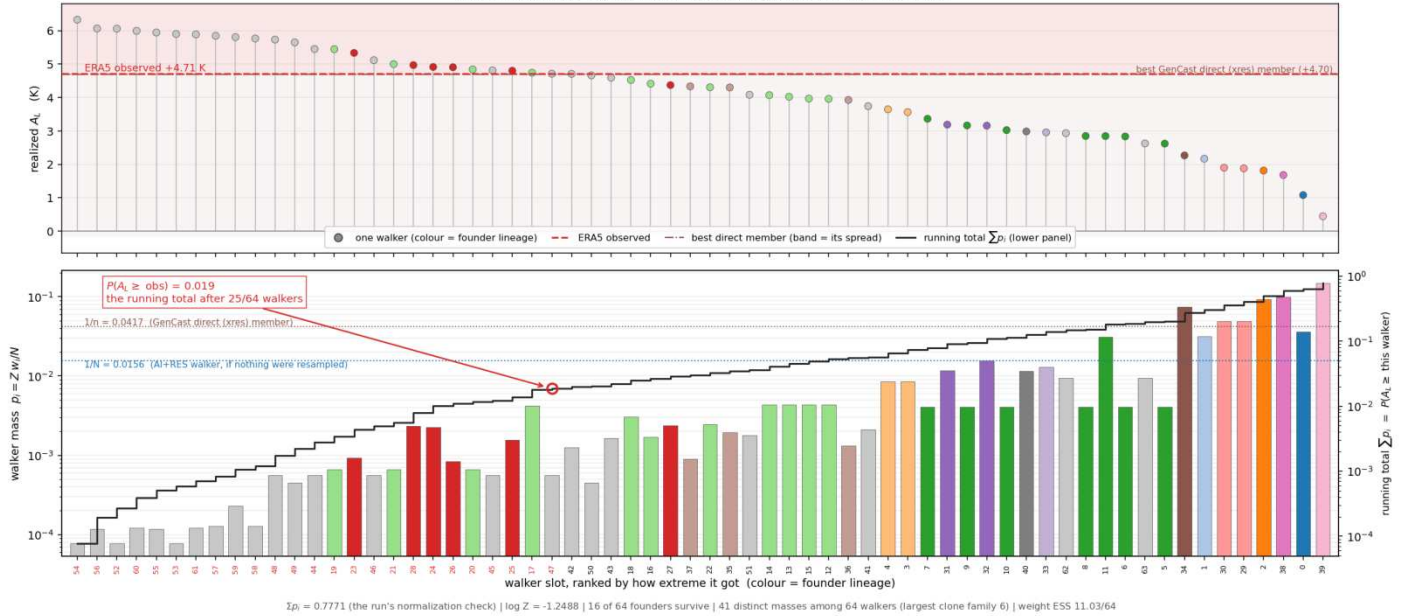
AI+RES pilot - California_HeatWave_2022 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 111 lineages killed, 16/64 founders still represented at the peak



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.

California Heat Wave 2022 (pilot) - all 64 surviving walkers, ranked hottest first



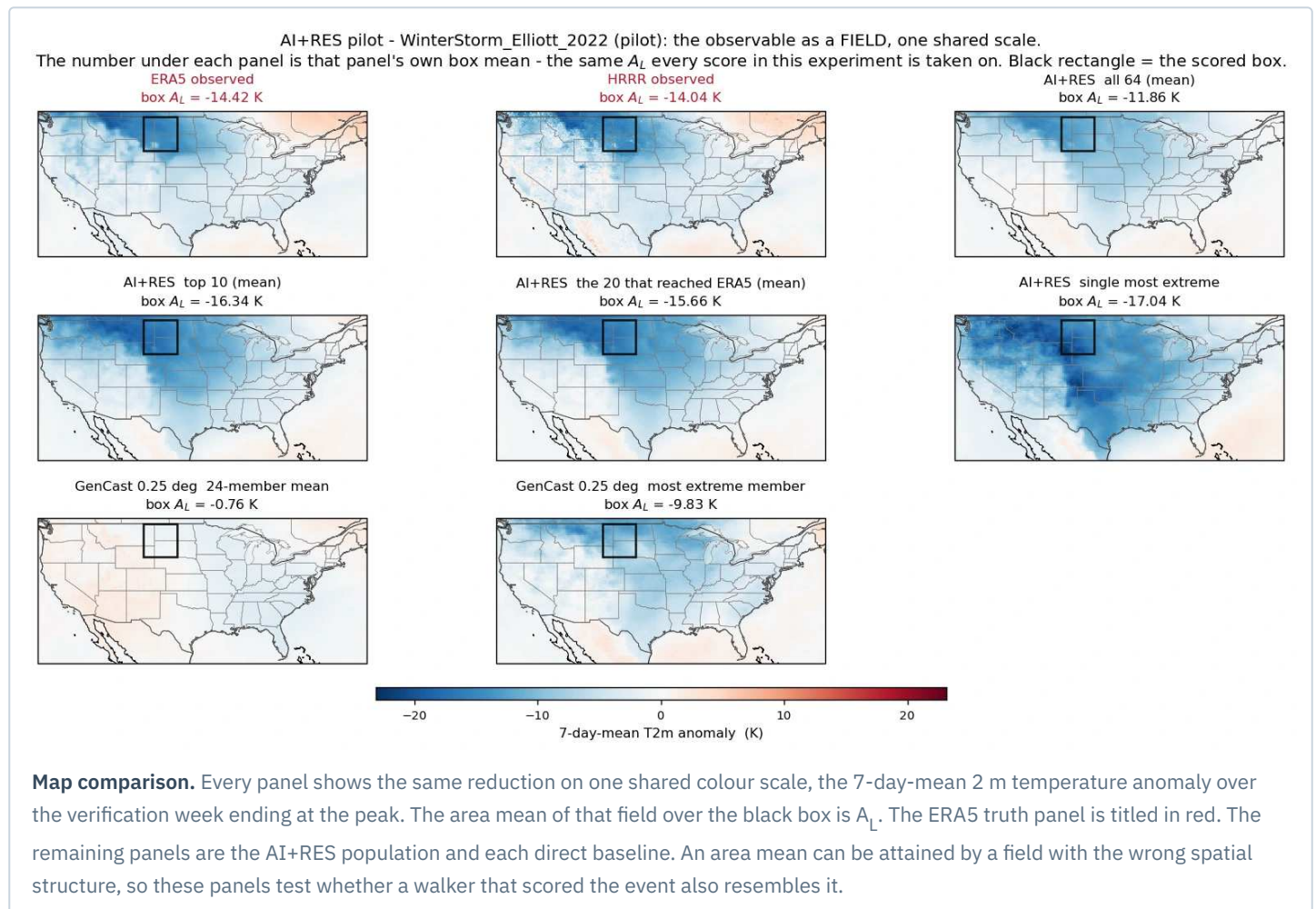
Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

The most extreme walker of the experiment, at -17.04 K, 2.6 K beyond the observation. This is the deeper of the two cold cases. It lies 0.5σ less deep in the model's distribution than the Southwest heat wave, which was missed, and it is reached.

OBSERVED A_L -14.42 K ERA5, 43-49 N, 106-100 W
CLIMATOLOGICAL PERCENTILE 98.41 th ERA5 1959-2022 · 99.13th against 1990-2019
RETURN PERIOD 5 yr 11 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD +2.81 σ week-3 ensemble -0.76 ± 4.87 K
DIRECT SAMPLING 0 / 24 best member -9.83 K · $P = 0.000$ [0.000, 0.142]
AI+RES 20 / 64 most extreme -17.04 K
WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$ 0.0079 sum of the walker probabilities past the observation
PEAK-DAY ANOMALY -19.71 K the single day, not the week · 99.0th percentile
64-YEAR RECORD FOR THIS SEASON -21.35 K (1983-12-25) peak 2022-12-23 · init 2022-12-02 (21 d lead)

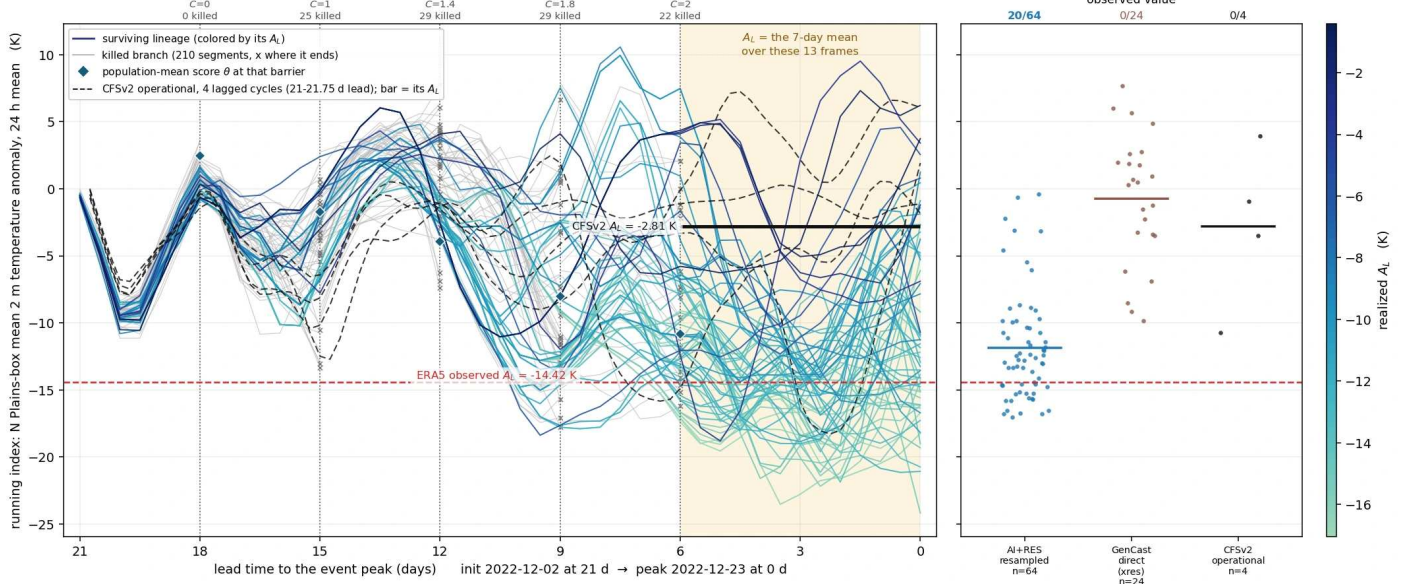
box: Northern Plains (43-49 N, 106-100 W) · registry key WinterStorm_Elliott_2022 · Slurm job 1184, 7 h 26 m
run health: log Z -1.1612 · normalization check 0.8108, the closest to 1 of the wave · 14/64
founders · ESS/N 1.00 / 0.39 / 0.29 / 0.33 / 0.59 · weight ESS 8.61

The run records the CONUS index alongside the box index at no extra GPU cost, which gives a second calibration anchor on the cold side at -2.73 K. There direct sampling returns $1/24 = 0.042$ [0.007, 0.202] and AI+RES returns 0.136, inside the interval. The weights undo the tilt only in part, taking an unweighted mean of -11.86 K to a weighted -3.64 K against a direct mean of -0.76 K. That is the pattern across the wave, in that the deeper the tilt, the less of it the reweighting recovers.

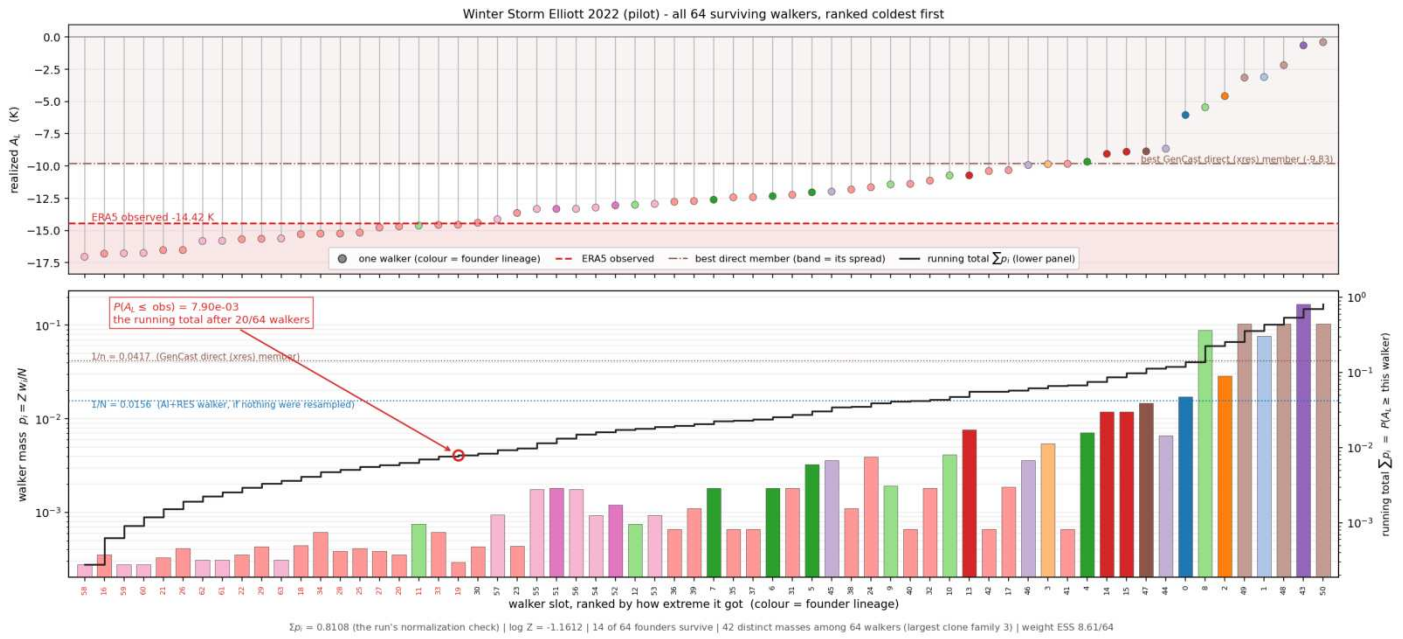


AI+RES pilot - WinterStorm_Elliott_2022 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 105 lineages killed, 14/64 founders still represented at the peak



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

Winter Storm Uri

February 2021

RUNG 4 OF 9

cold reached

The shallower of the two cold cases, and the case in which the weights recover the direct-sampling mean most closely. The schedule was carried unchanged from four heat events to a Texas freeze, and the run reaches the observation and assigns it a probability.

OBSERVED A_L

-7.40 K ERA5, 26-37 N, 106-94 W

CLIMATOLOGICAL PERCENTILE

96.80 ERA5 1959-2022 · 98.45th against 1990-2019

RETURN PERIOD

4 yr 19 of 64 years hold a window this extreme

DEPTH IN GENCAST'S SPREAD

+2.28 σ week-3 ensemble $+0.06 \pm 3.27$ K

DIRECT SAMPLING

0 / 24 best member -6.51 K · P = 0.000 [0.000, 0.142]

AI+RES

32 / 64 most extreme -10.20 K

WEIGHTED P(A_L BEYOND OBSERVED)

0.0090 sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

-19.95 K the single day, not the week · 100.0th percentile

64-YEAR RECORD FOR THIS SEASON

-14.73 K (2021-02-20) peak 2021-02-15 · init 2021-01-25 (21 d lead)

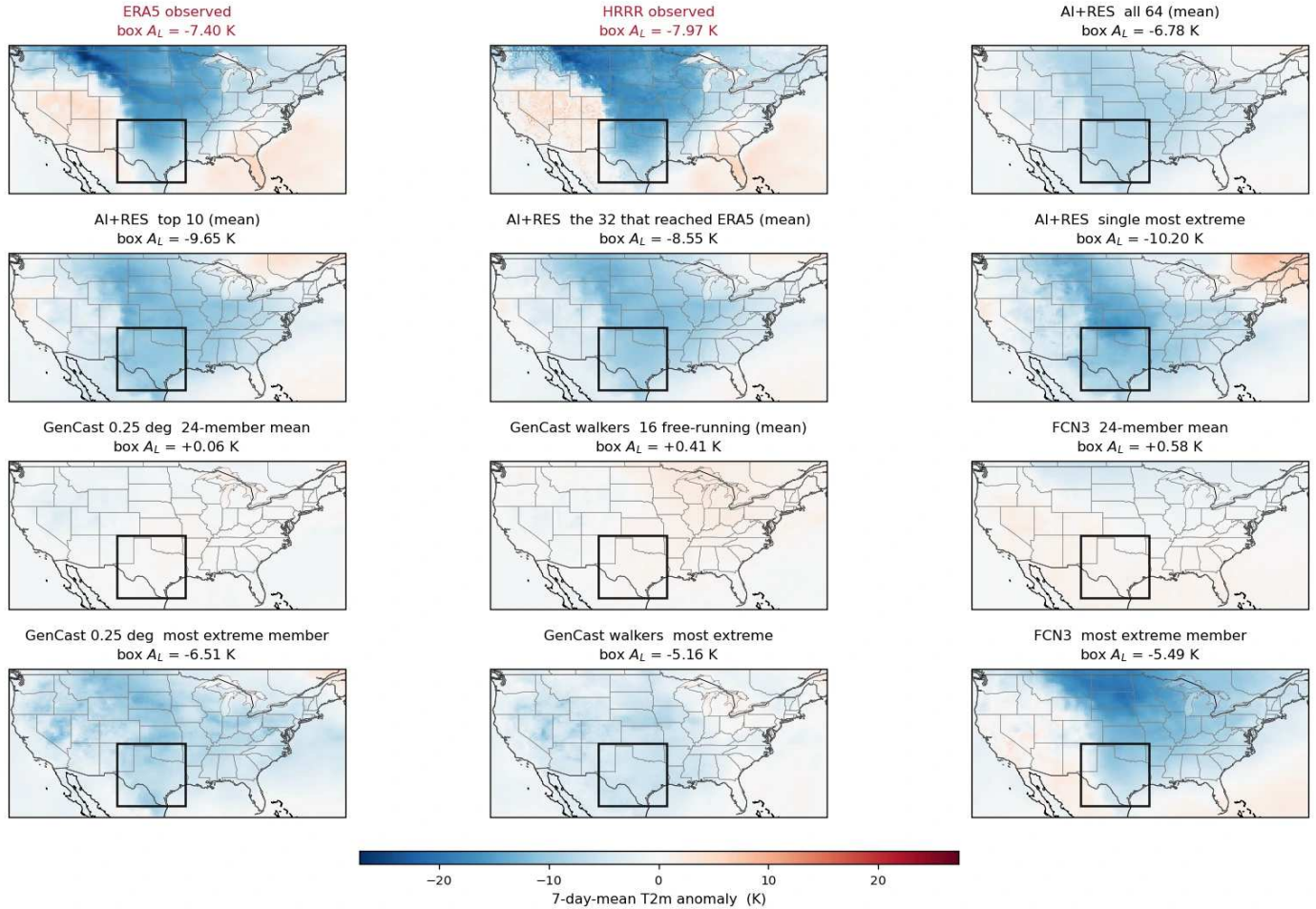
box: Texas and Oklahoma (26-37 N, 106-94 W) · registry key WinterStorm Uri_2021 · Slurm job 1183, 7 h 12 m
run health: log Z -0.6735 · normalization check 1.7688 · 16/64 founders · ESS/N 1.00 / 0.40 / 0.43 / 0.41 / 0.30 · weight ESS only 2.24 of 64, so read the spatial map as shape, not level

Three independent direct baselines hold no member beyond the observation, namely 16 free-running walkers, the 24-member GenCast cube and the 24-member FCN3 ensemble, with the best member 0.9 K short. The weights undo the tilt. An unweighted mean of -6.78 K reweights to +0.21 K against a direct mean of +0.41 K, and the population still covers the bulk of the distribution, its range reaching +1.57 K. The weighted and direct curves agree over the range direct sampling resolves, with the weighted curve running slightly low, and the weighted curve then extends about two decades past the point where every direct curve stops.

The 4-year return period is an artefact of the observable rather than a statement about the severity of the storm. A 7-day box mean over 26-37 N averages a 3-day collapse across eleven degrees of latitude. The single peak day was -19.95 K, the most extreme mid-February day in the 64-year record for that box. Uri did its damage through duration and through minimum temperature in a region without winterisation, and A_L measures neither.

AI+RES pilot - WinterStorm_Uri_2021 (pilot): the observable as a FIELD, one shared scale.

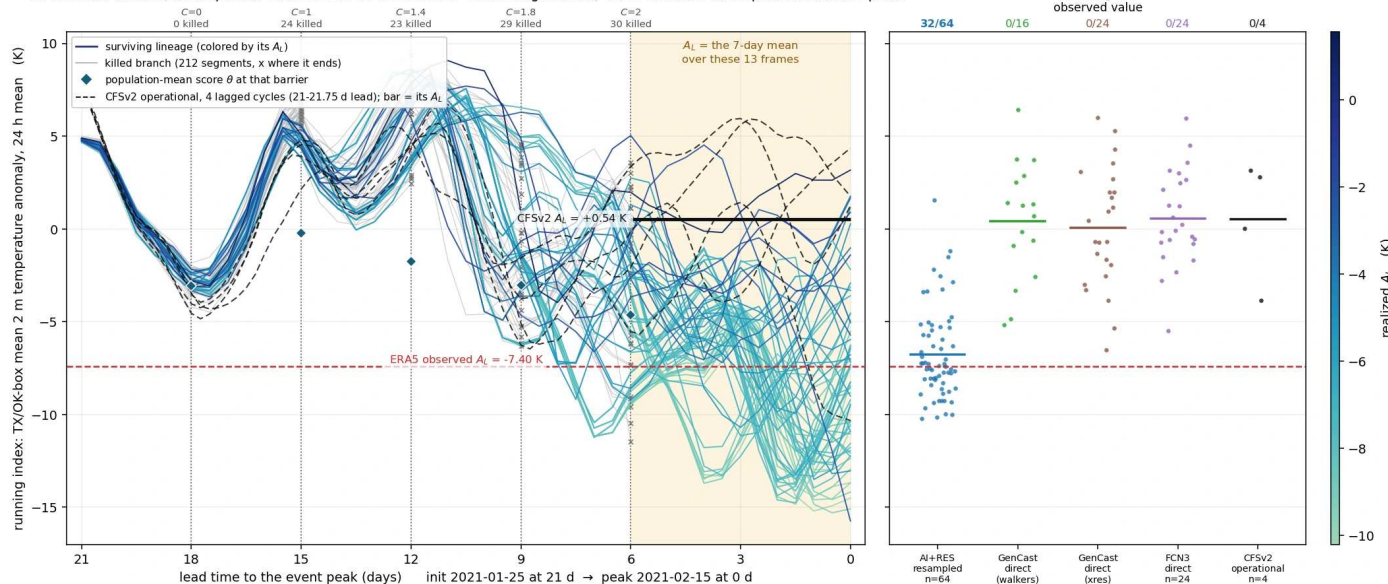
The number under each panel is that panel's own box mean - the same A_L every score in this experiment is taken on. Black rectangle = the scored box.



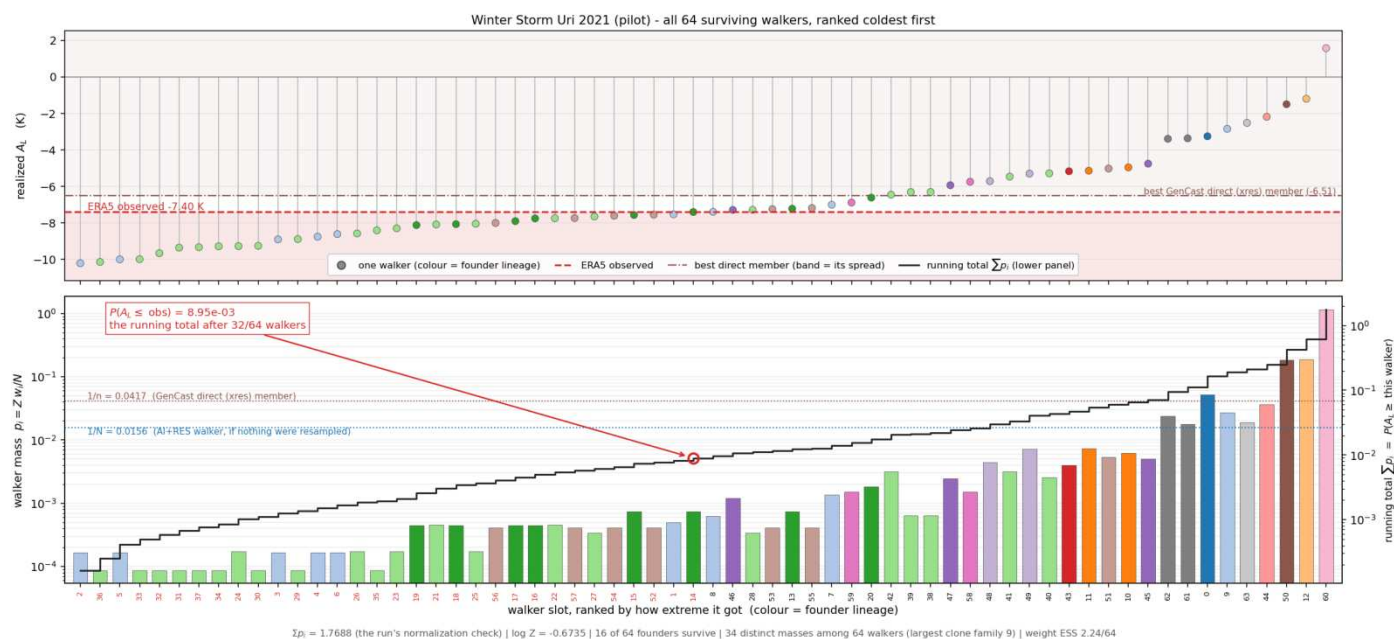
Map comparison. Every panel shows the same reduction on one shared colour scale, the 7-day-mean 2 m temperature anomaly over the verification week ending at the peak. The area mean of that field over the black box is A_L . The ERA5 truth panel is titled in red. The remaining panels are the AI+RES population and each direct baseline. An area mean can be attained by a field with the wrong spatial structure, so these panels test whether a walker that scored the event also resembles it.

AI+RES pilot - WinterStorm_Uri_2021 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 106 lineages killed, 16/64 founders still represented at the peak



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.

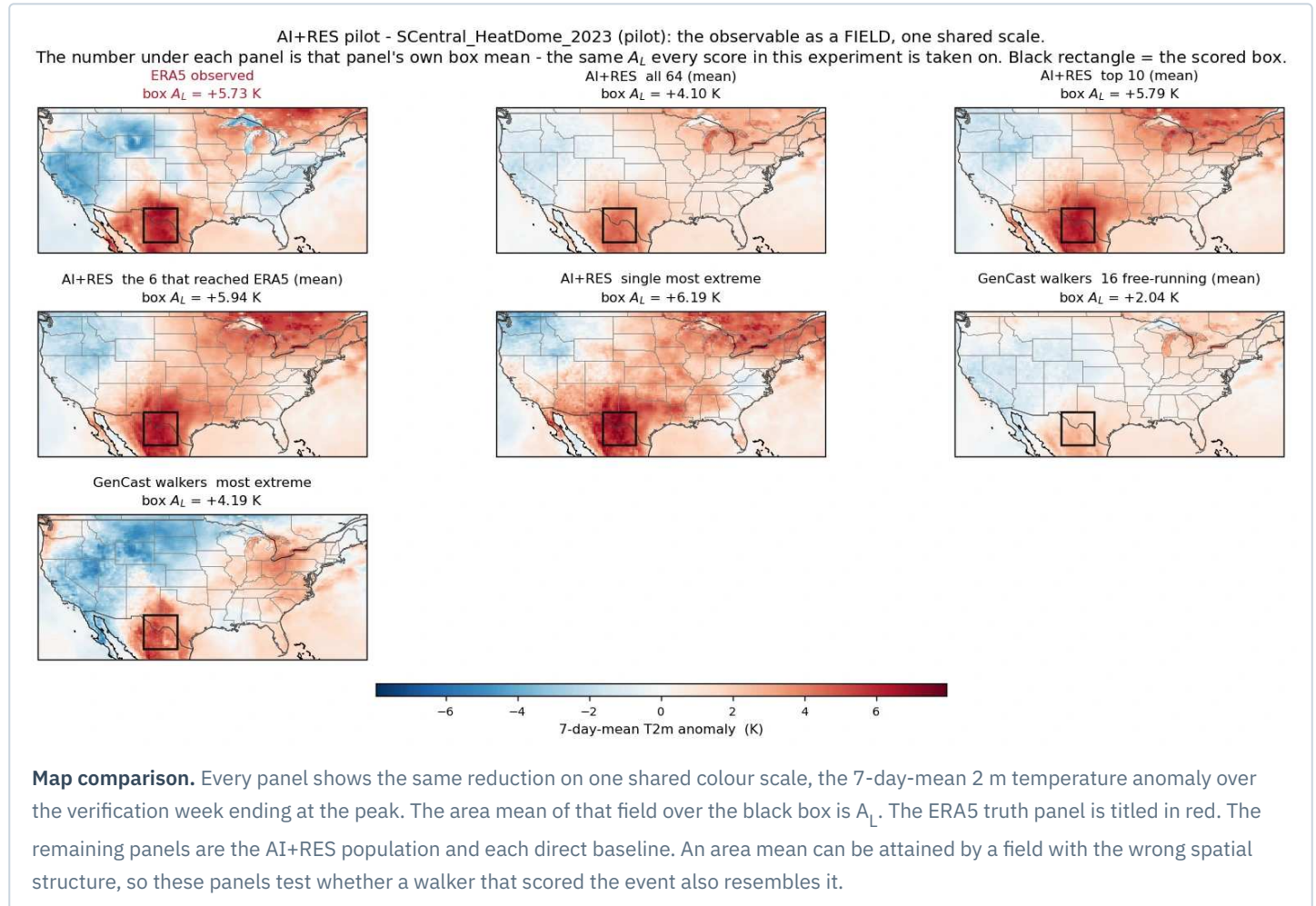


Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

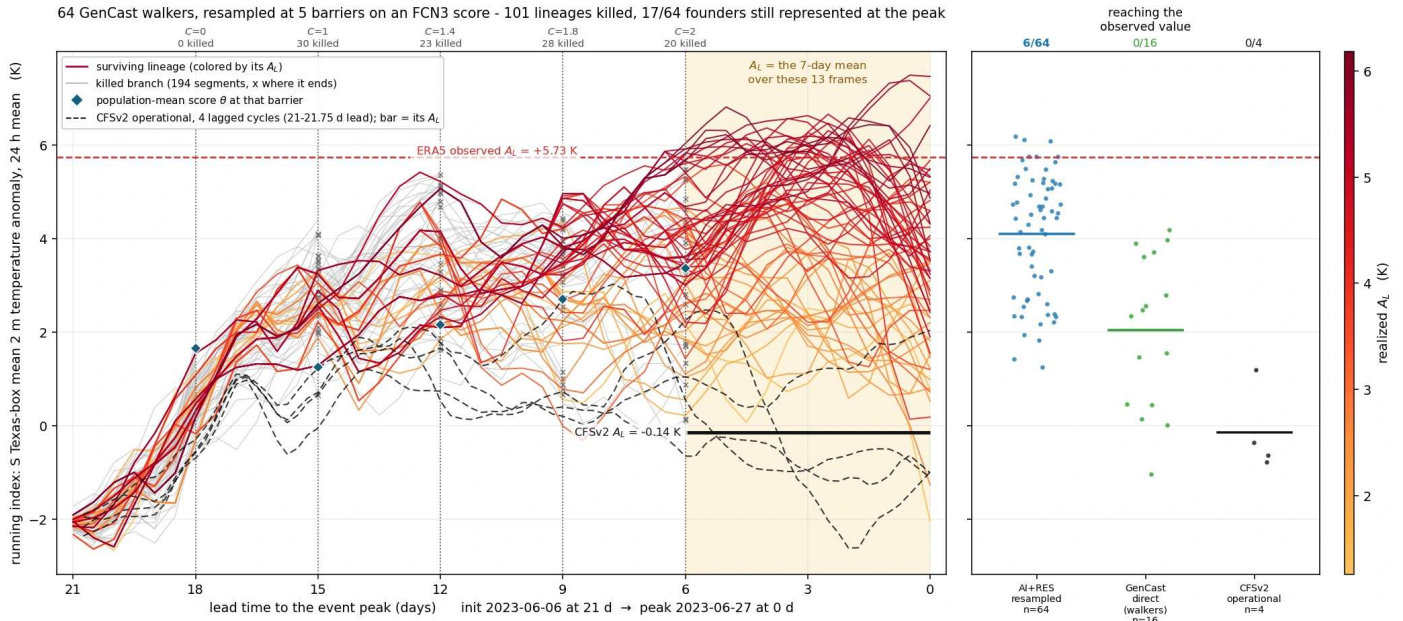
The rarest case on the slate against its own climatology. No window in 64 years of late-June Texas comes within 0.75 K of the observation. The schedule was applied unchanged in a subtropical-ridge regime, and the run reaches the observation.

OBSERVED A_L +5.73 KERA5, 26-32 N, 106-100 W
CLIMATOLOGICAL PERCENTILE 100.00thERA5 1959-2022 · 100.00th against 1990-2019
RETURN PERIOD beyond the record0 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD ~+2.2 σweek-3 ensemble not tabulated
DIRECT SAMPLING 0 / 16best member +4.19 K · P = 0.000 [0.000, 0.206]
AI+RES 6 / 64most extreme +6.19 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.00273sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +5.54 Kthe single day, not the week · 100.0th percentile
64-YEAR RECORD FOR THIS SEASON +4.98 K (2011-06-21)peak 2023-06-27 · init 2023-06-06 (21 d lead)

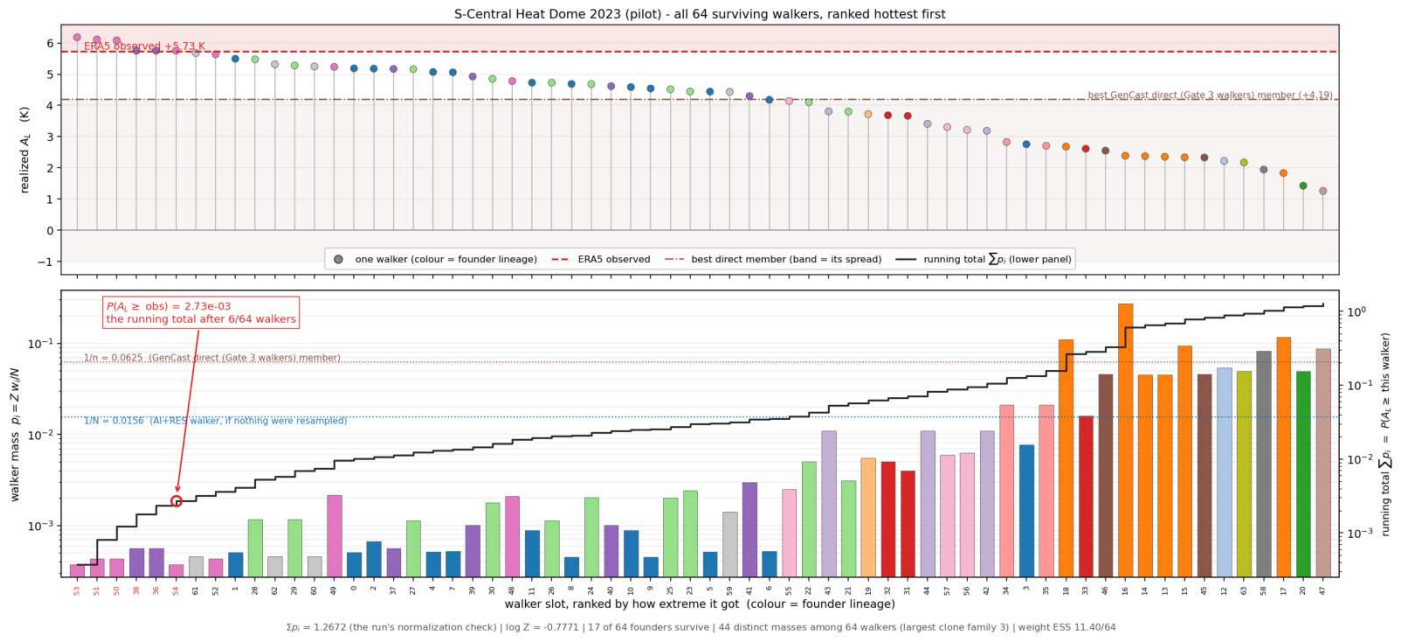
The C schedule was tuned once, on the Pacific Northwest case, and has not been re-tuned since. Reaching the observation in a different regime is therefore evidence that the schedule transfers, rather than evidence that it was fitted to this case. The weights recover the direct mean, giving a weighted mean of +2.35 K against a direct mean of +2.04 K. The core of the dome extended into Mexico, south of the 24 N crop, so this box covers the northern part of the event, as the Pacific Northwest box does.



AI+RES pilot - S-Central_HeatDome_2023 (pilot): the walk itself, barrier by barrier



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

Pacific Northwest Heat Dome

June 2021

RUNG 6 OF 9

heat reached

The pilot case, against which the other runs are compared. Forty direct GenCast members initialised three weeks ahead do not reach the dome. 42 of 64 walkers do.

OBSERVED A_L

+7.72 K ERA5, 44-50 N, 124-118 W

CLIMATOLOGICAL PERCENTILE

99.60 ERA5 1959-2022 · 99.61th against 1990-2019

RETURN PERIOD

22 yr 3 of 64 (1992, 2015, 2021) years hold a window this extreme

DEPTH IN GENCAST'S SPREAD

+1.98 σ week-3 ensemble $+2.46 \pm 2.65$ K

DIRECT SAMPLING

0 / 40 best member $+7.49$ K · $P = 0.000$ [0.000, 0.088]

AI+RES

42 / 64 most extreme $+10.29$ K

WEIGHTED $P(A_L \text{ BEYOND OBSERVED})$

0.054 sum of the walker probabilities past the observation

PEAK-DAY ANOMALY

+14.38 K the single day, not the week · 100.0th percentile

64-YEAR RECORD FOR THIS SEASON

+11.11 K - the same dome, 3 d later peak 2021-06-28 · init 2021-06-07 (21 d lead)

box: Oregon and Washington, the dome core (44-50 N, 124-118 W) · registry key PNW_HeatDome_2021

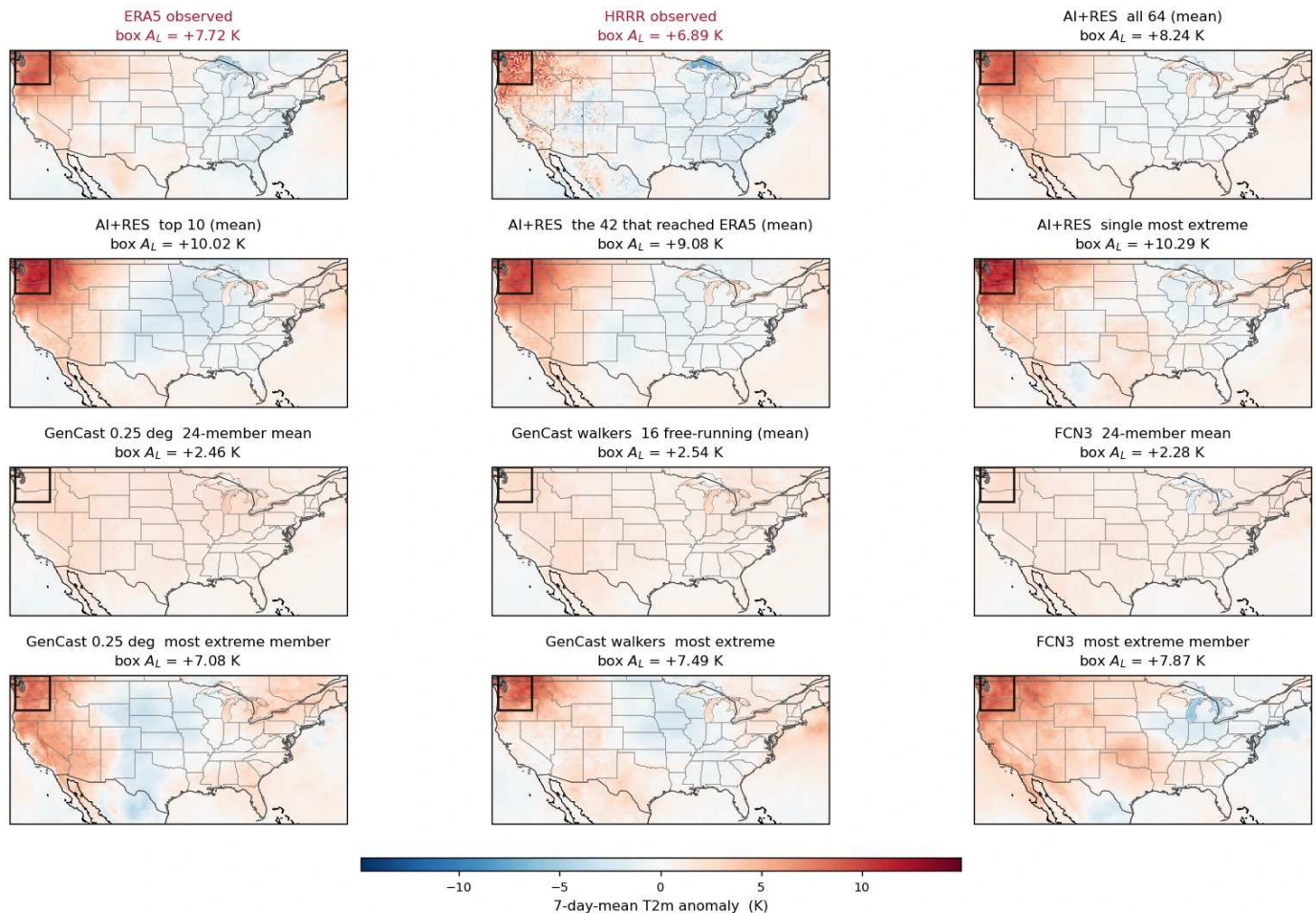
· Slurm job 1172, 7 h 45 m

run health: log Z -0.9317 · normalization check 0.5684 · 11/64 founders · weight ESS 5.68

The observed maximum reached about 52 N into British Columbia, outside the 50 N CONUS crop, so this box covers the southern two thirds of the event. On the per-grid-point tail the weighted spatial distribution gives $P(\text{point} \geq +9 \text{ K}) = 0.0505$, against 0.0848 in ERA5 and 0.0011 from 24 direct members.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the observable as a FIELD, one shared scale.

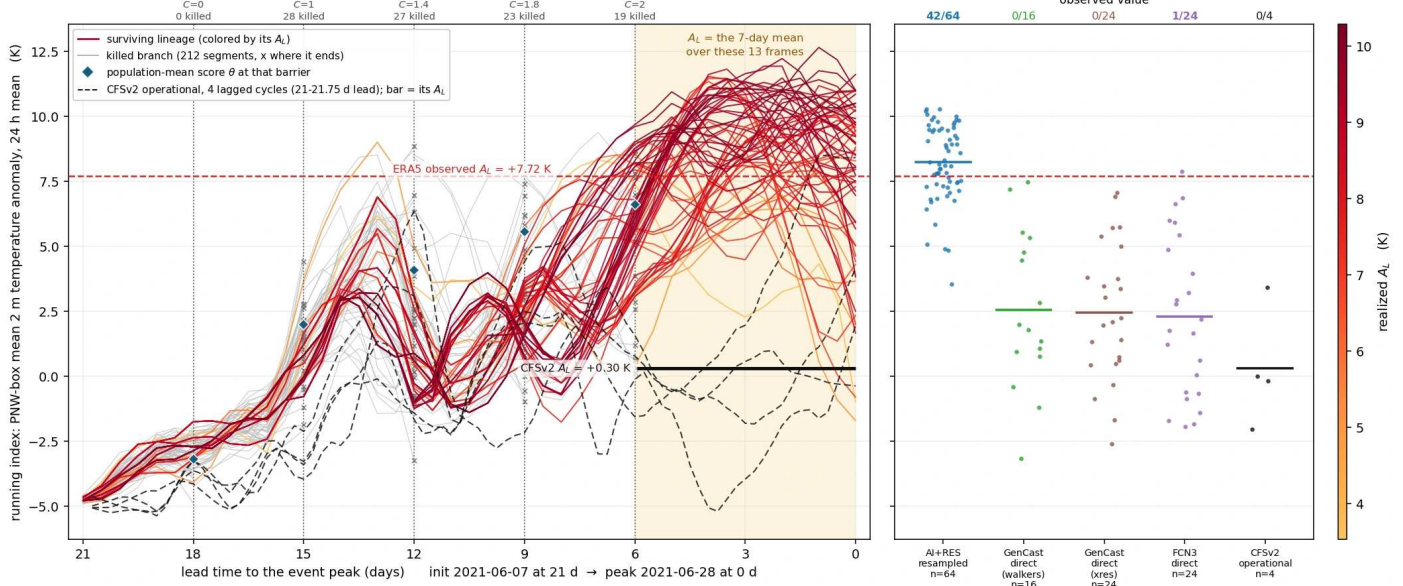
The number under each panel is that panel's own box mean - the same A_L every score in this experiment is taken on. Black rectangle = the scored box.



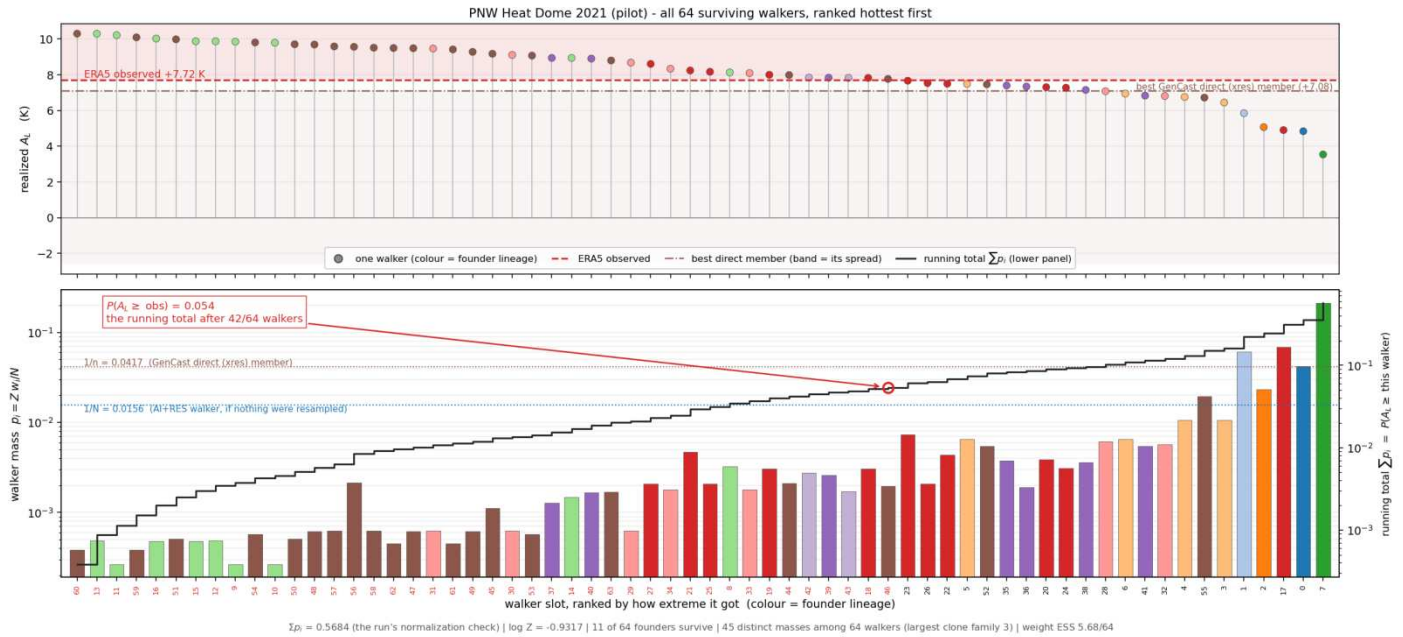
Map comparison. Every panel shows the same reduction on one shared colour scale, the 7-day-mean 2 m temperature anomaly over the verification week ending at the peak. The area mean of that field over the black box is A_L . The ERA5 truth panel is titled in red. The remaining panels are the AI+RES population and each direct baseline. An area mean can be attained by a field with the wrong spatial structure, so these panels test whether a walker that scored the event also resembles it.

AI+RES pilot - PNW_HeatDome_2021 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 97 lineages killed, 11/64 founders still represented at the peak



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

The Nationwide Warm Spell

CONUS index

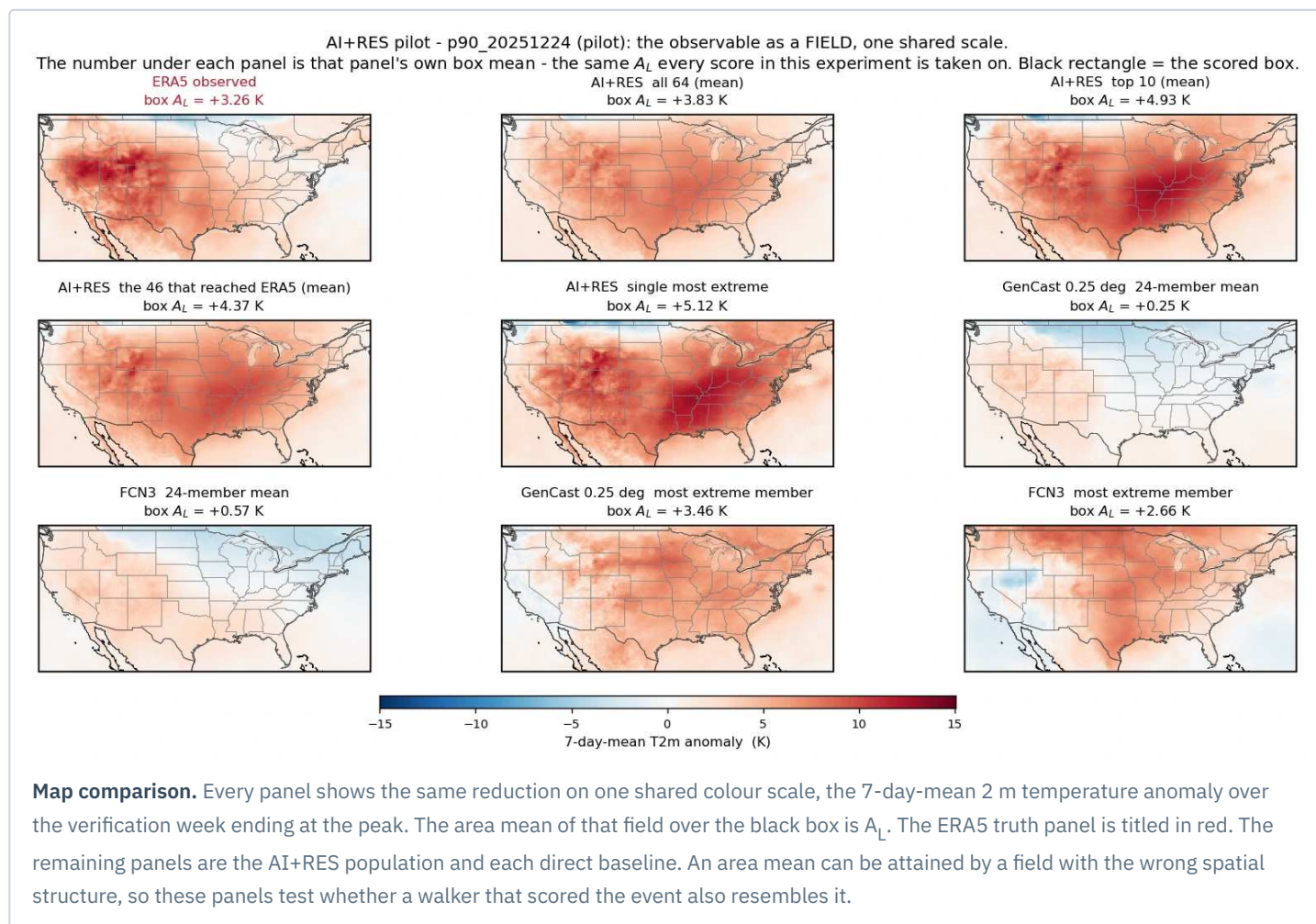
calibration anchor

The one rung on which direct sampling returns a non-zero probability, so the weighted estimate can be checked against it. The weighted estimate falls inside the direct interval.

OBSERVED A_L +3.26 KERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 98.78thERA5 1959-2022 · 97.40th against 1990-2019
RETURN PERIOD 10 yr8 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD +1.91 σweek-3 ensemble +0.25 ± 1.58 K
DIRECT SAMPLING 1 / 24best member +3.46 K · P = 0.042 [0.001, 0.211]
AI+RES 46 / 64most extreme +5.12 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.0661sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +4.91 Kthe single day, not the week · 99.8th percentile
64-YEAR RECORD FOR THIS SEASON +4.45 K (2015-12-15)peak 2025-12-24 · init 2025-12-03 (21 d lead)

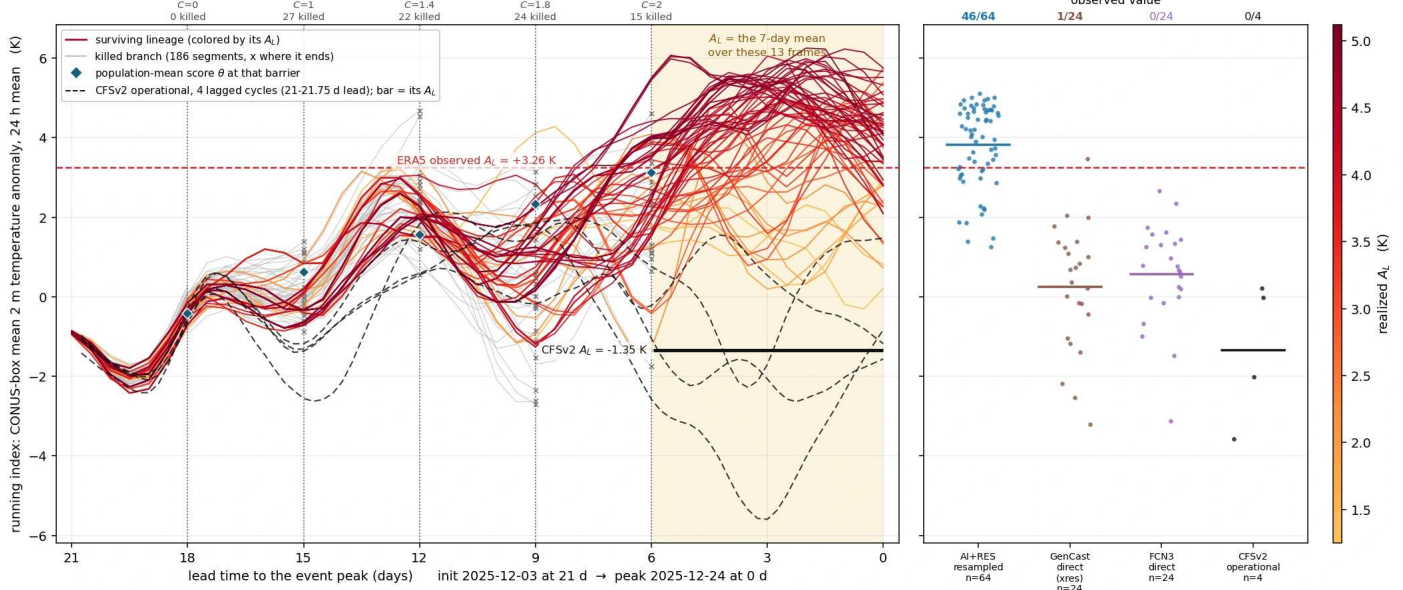
box: Whole CONUS - this case is defined on the national mean (24-50 N, 125-66 W) · registry key p90_20251224 · Slurm job 1182, 8 h 42 m
run health: log Z -1.1846 · normalization check 0.6890 · 16/64 founders · ESS/N 1.00 / 0.32 / 0.56 / 0.54 / 0.69, the only run that recovers after every barrier

AI+RES gives 0.0661 against direct sampling's $1/24 = 0.0417$ [0.0074, 0.202]. No wave-1 run supports this comparison, because each had direct sampling at 0/24 and the weighted probability had nothing to be compared with. Through the mid-range the weighted curve runs about twice the direct estimate, a discrepancy of 2.2 to 2.6 binomial standard errors at $n = 24$. The weighted mean cannot return to the direct mean in this case, because resampling killed every lineage below +1.25 K. That is a limit of the support of the surviving population, not a weighting error.

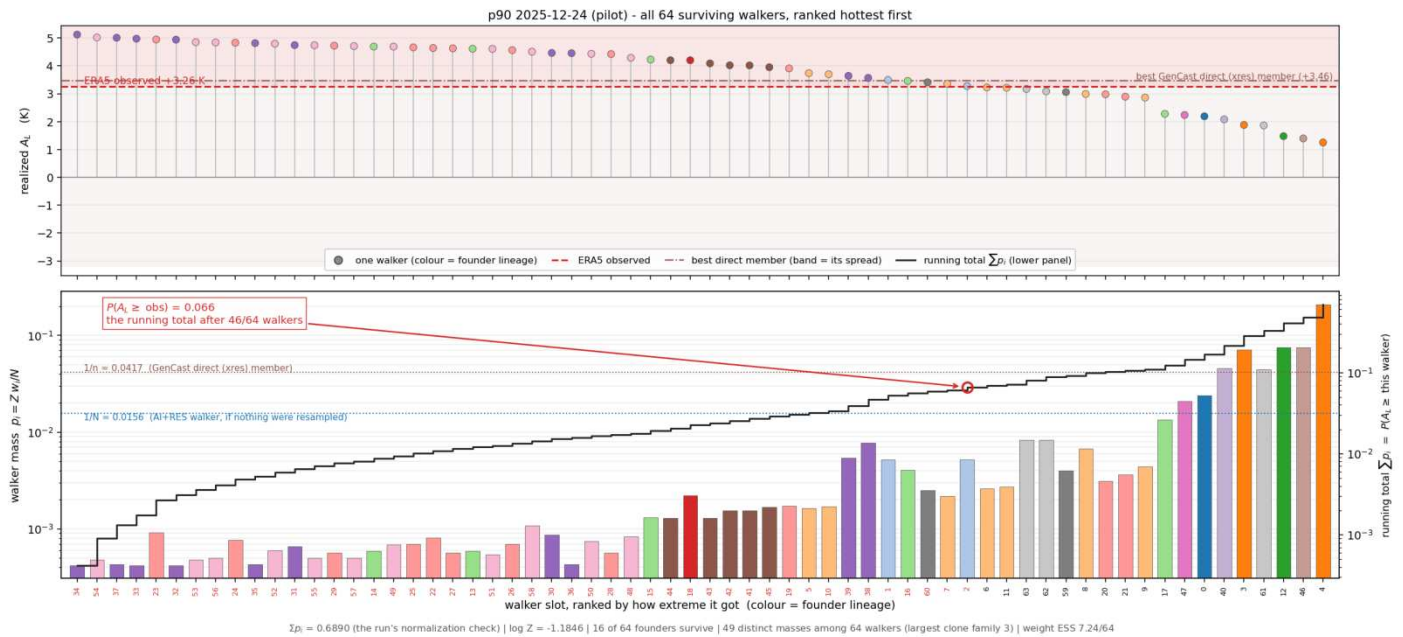


AI+RES pilot - p90_20251224 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 88 lineages killed, 16/64 founders still represented at the peak



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



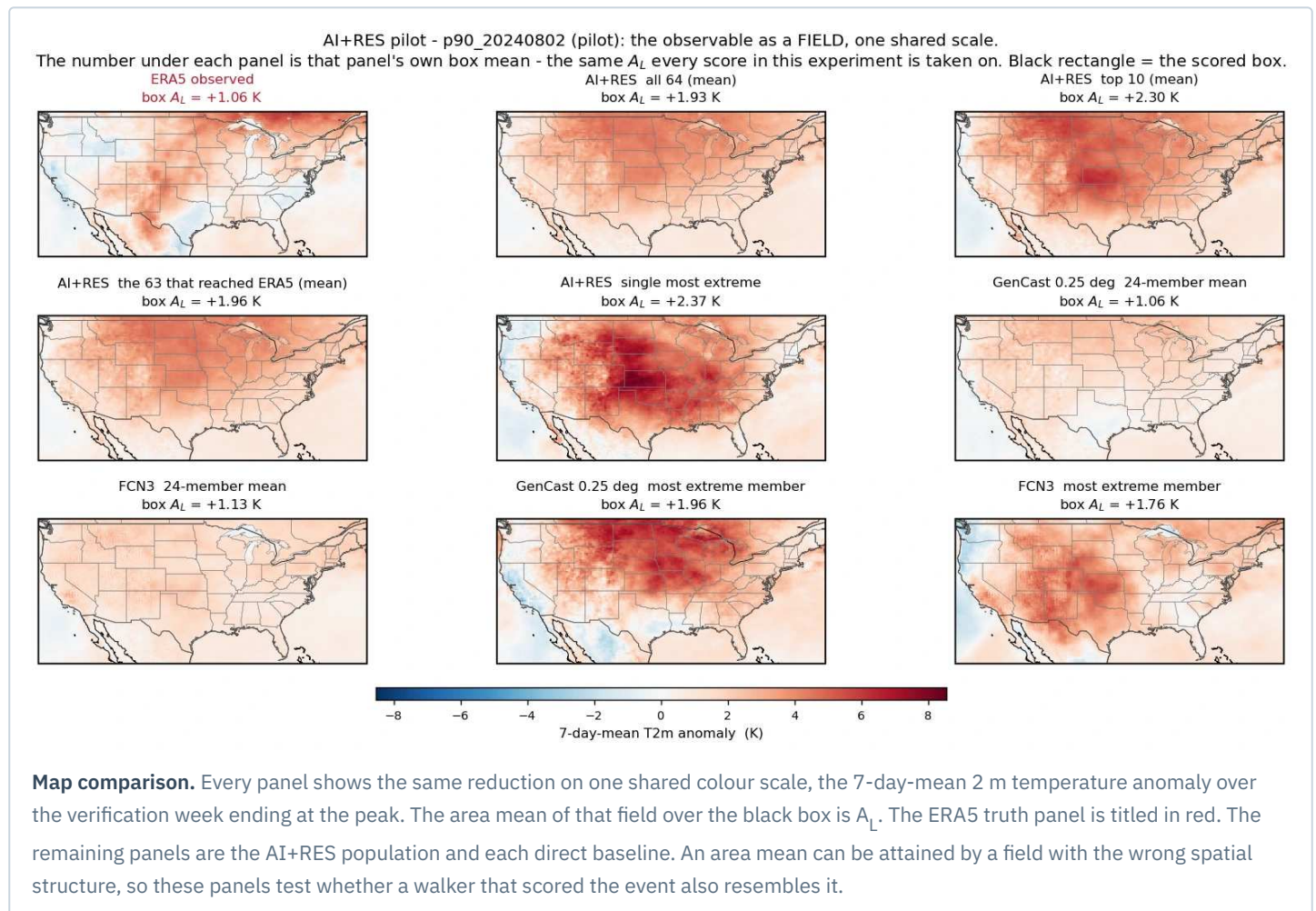
Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

The no-harm control. The observation lies at the ensemble median, so a sampler built to push walkers into the tail should still return a probability near one half. It returns 0.649 against direct sampling's 0.583.

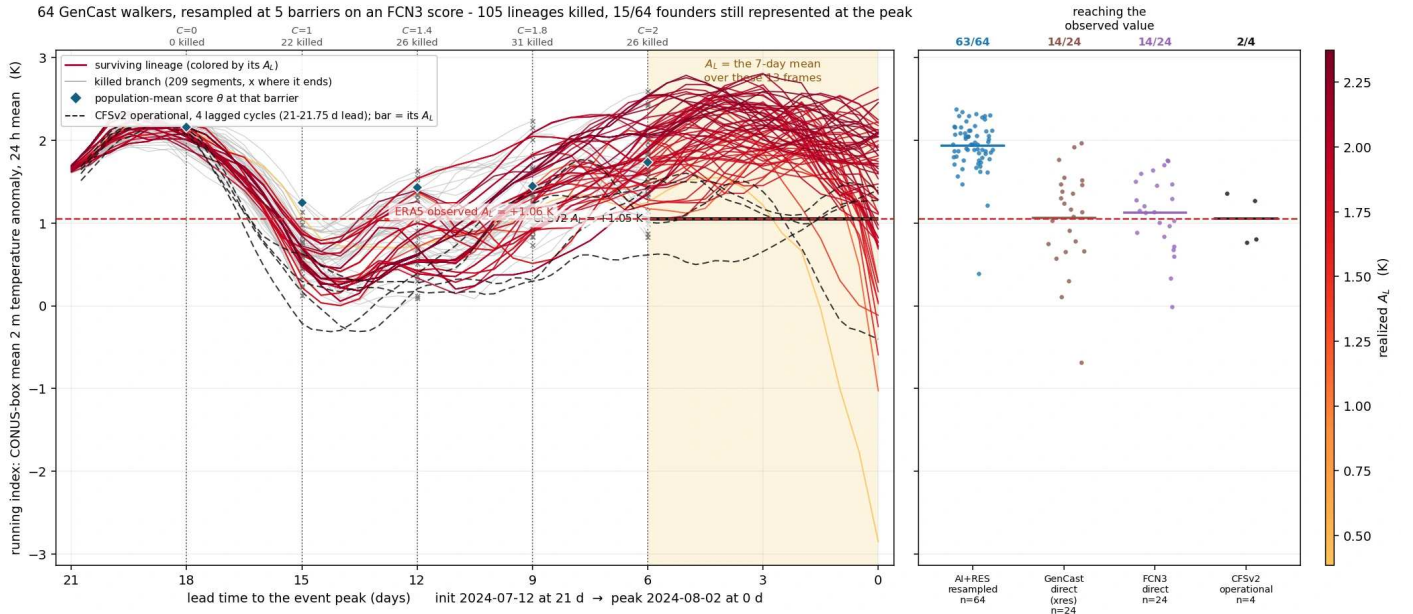
OBSERVED A_L +1.06 KERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 94.08thERA5 1959-2022 · 91.55th against 1990-2019
RETURN PERIOD 4 yr20 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD 0.00 σweek-3 ensemble +1.06 ± 0.60 K
DIRECT SAMPLING 14 / 24best member +1.96 K · P = 0.583 [0.366, 0.779]
AI+RES 63 / 64most extreme +2.37 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.649sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +2.40 Kthe single day, not the week · 100.0th percentile
64-YEAR RECORD FOR THIS SEASON +2.06 K (2006-07-20)peak 2024-08-02 · init 2024-07-12 (21 d lead)

box: Whole CONUS - this case is defined on the national mean (24-50 N, 125-66 W) · registry key p90_20240802 · Slurm job 1185, 7 h 48 m
run health: log Z -0.6063 · normalization check 0.7039 · 15/64 founders · ESS/N 1.00 / 0.53 / 0.37 / 0.30 / 0.40

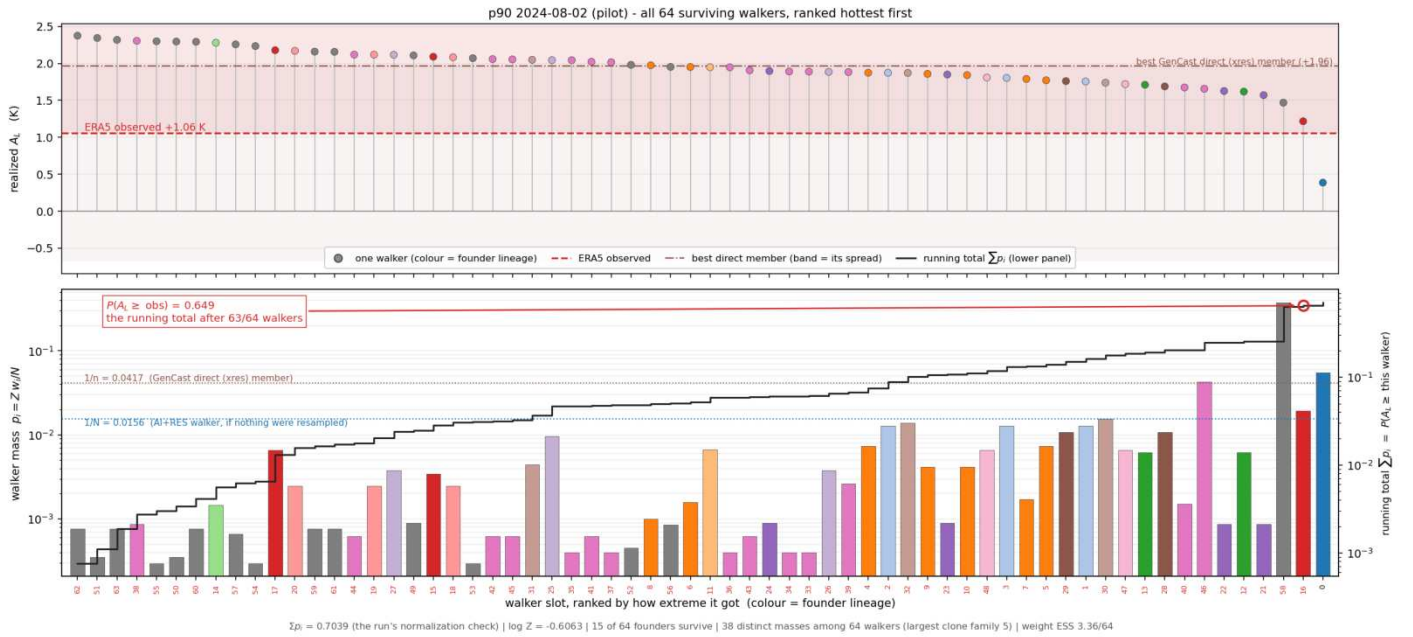
The reach count is not a result. 63 of 64 walkers reach a median target, which reports where the tilt pushed the population and nothing more. The test is whether the estimator returns a small probability, near 0.05, for a median target, and it does not. On the per-grid-point tail the weighted population gives 0.0727 against 0.0702 for direct GenCast, an agreement to 4 %, while the unweighted population gives 0.1627, heavier by a factor of 2.3.



AI+RES pilot - p90_20240802 (pilot): the walk itself, barrier by barrier



Trajectory. A_L against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



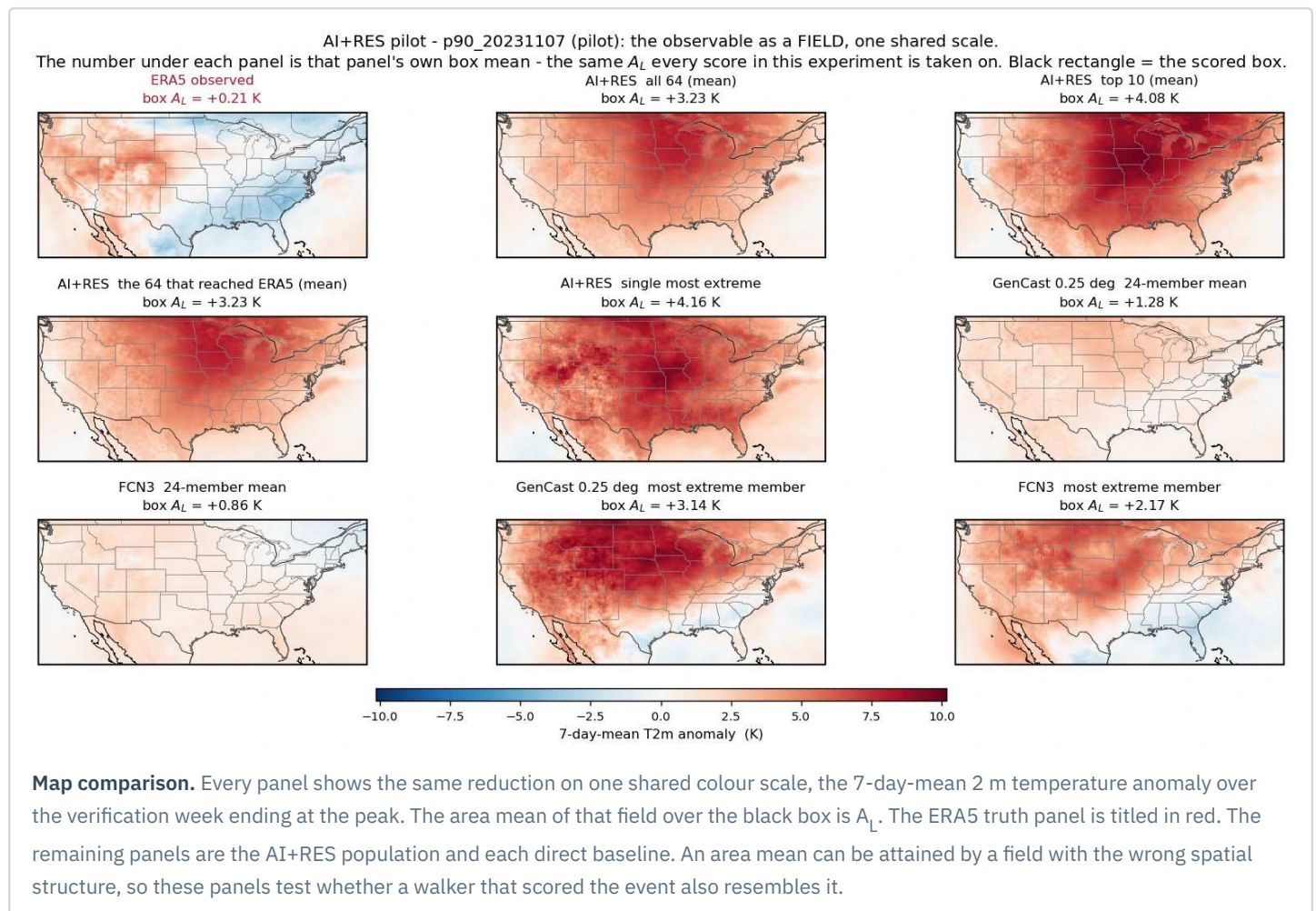
Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_L , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

The estimator degenerates at this target. Direct sampling puts this November week at 0.875. AI+RES reports 0.583, below the direct interval, and cannot report any other value.

OBSERVED A_L +0.21 KERA5, 24-50 N, 125-66 W
CLIMATOLOGICAL PERCENTILE 60.10thERA5 1959-2022 · 53.64th against 1990-2019
RETURN PERIOD 1 yr62 of 64 years hold a window this extreme
DEPTH IN GENCAST'S SPREAD -0.97 σweek-3 ensemble +1.28 ± 1.09 K
DIRECT SAMPLING 21 / 24best member +3.14 K · P = 0.875 [0.676, 0.973]
AI+RES 64 / 64most extreme +4.16 K
WEIGHTED P(A_L BEYOND OBSERVED) 0.583 (pinned)sum of the walker probabilities past the observation
PEAK-DAY ANOMALY +3.01 Kthe single day, not the week · 98.7th percentile
64-YEAR RECORD FOR THIS SEASON +3.38 K (2020-11-10)peak 2023-11-07 · init 2023-10-17 (21 d lead)

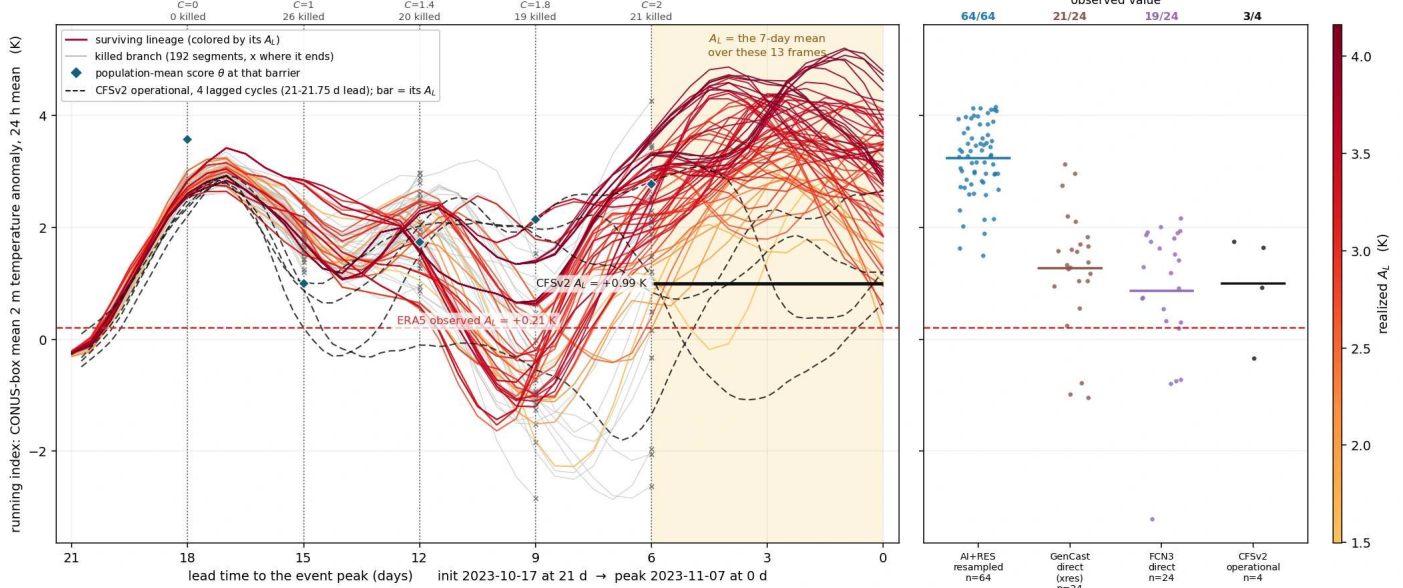
The cause is structural and was checked numerically. Resampling tilted the whole population warm, so its lowest realized A_L is +1.498 K, above the +0.211 K target, and the indicator $1\{A_L \geq \text{observed}\}$ equals 1 for all 64 walkers. The estimate $Z \cdot \text{mean}(w \cdot 1\{\dots\})$ then reduces to $Z \cdot \text{mean}(w)$, which is the normalization check, 0.582662 against 0.582662 to every digit printed. The estimator cannot return a probability above its own normalization check, and the observed value lies above that ceiling. Self-normalizing does not help, since it returns 1.000, the only information the population retains being that every walker reached the target. Nothing inside the run reports this condition, and a reader of compare.json alone would take 0.583 at face value.

AI+RES therefore has no resolution at a target that the tilt has already carried the entire population past. The usable operating range is the set of targets that the resampled population still straddles. The Normal-Day Test case meets that condition, with a population minimum of +0.39 K against a +1.06 K target. This case does not.

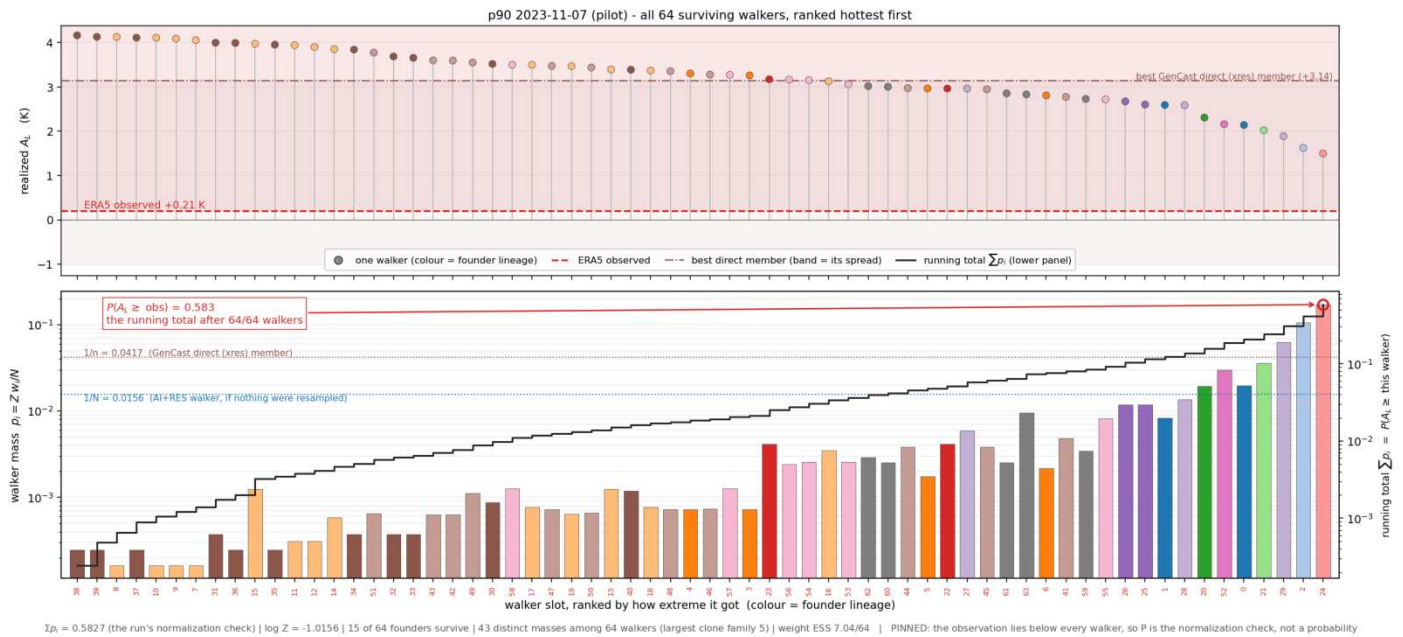


AI+RES pilot - p90_20231107 (pilot): the walk itself, barrier by barrier

64 GenCast walkers, resampled at 5 barriers on an FCN3 score - 86 lineages killed, 15/64 founders still represented at the peak



Trajectory. A_t against lead time for every branch the run rolled. Killed lineages are grey and the 64 surviving lineages are coloured. Vertical lines mark the resampling barriers, the shaded interval is the 7-day verification window, the observed value is red, and the operational CFSv2 forecast at the same lead is black.



Per-walker anomaly and probability mass. The upper panel ranks all 64 surviving walkers by their A_t , against the observed value in red and the best direct-sampling member in brown. The lower panel gives the probability mass $p_i = Z \cdot w_i / N$ carried by each walker, on a log axis, with the running total $\sum p_i$ as the black staircase. The headline probability is the value of that staircase where the ranking crosses the observation. Two reference lines set the scale, $1/n$ for a single direct member and $1/N$ for a walker that had not been resampled. Bars of identical height are clones of one parent at the last resampling and carry identical mass by construction, so the number of distinct masses in the panel, not 64, is the number of independent pieces of information the estimate rests on.

The persistence control

same walkers, same schedule, different score

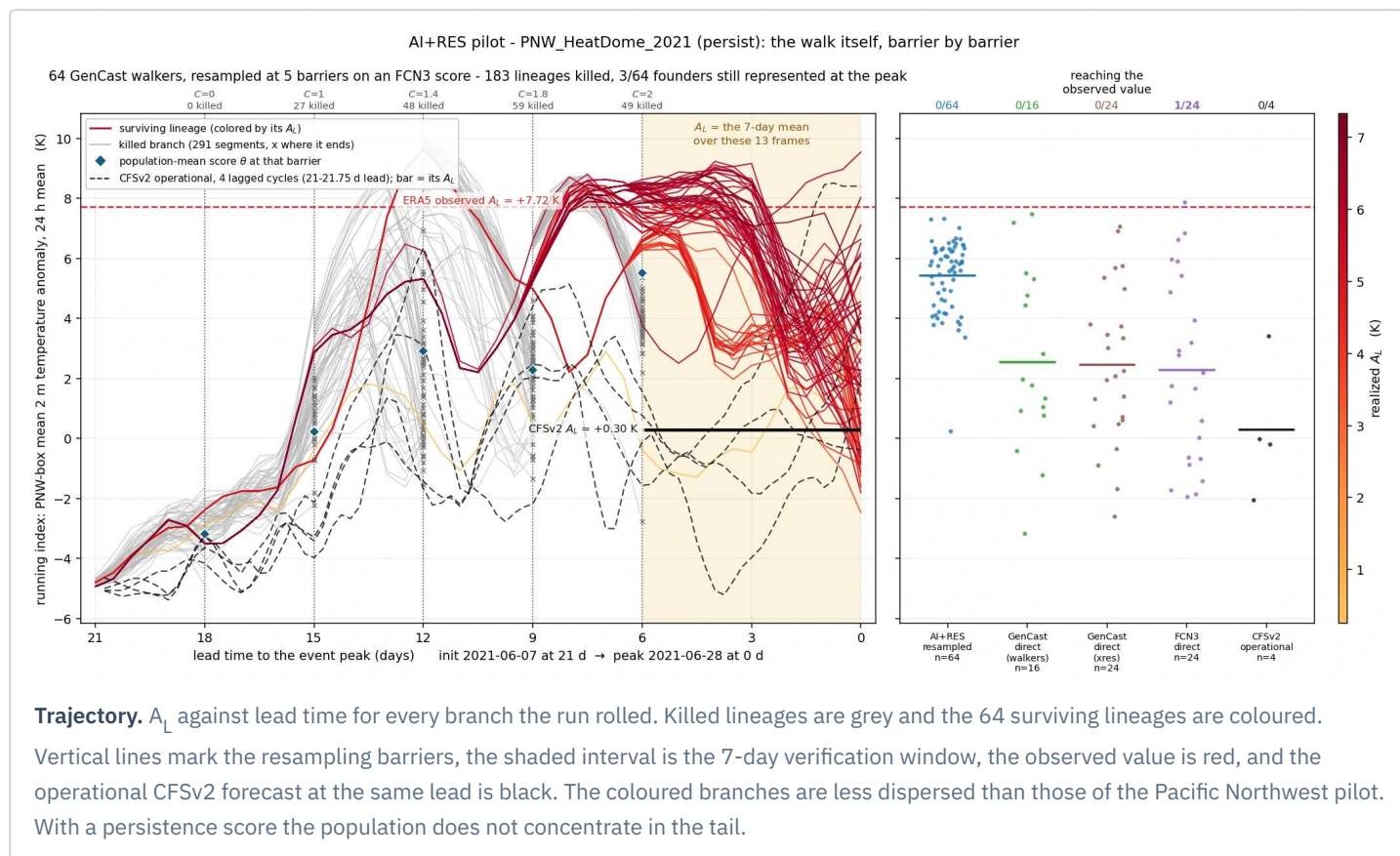
ATTRIBUTION

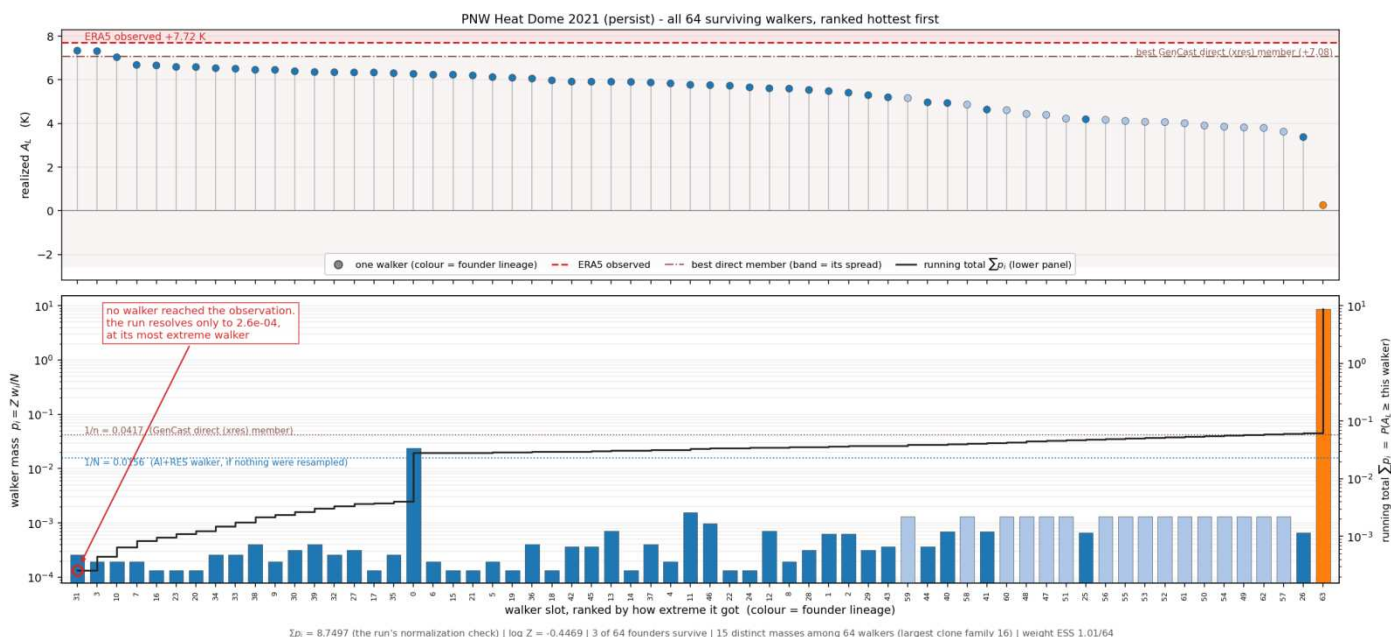
not reached

This run attributes the gains reported above to the FCN3 score. It repeats the Pacific Northwest pilot with the same walkers, the same resampling, the same C schedule and the same seeds, and changes only the function that decides which walkers to clone, from a forecast of the event to the current-state temperature. It reaches the observed +7.72 K with **0 of 64** walkers.

The selection carries no information at this lead. ESS/N falls to 0.04 and 0.02 at steps 3 and 4, with clone multiplicities of 39 and 49, three of 64 founders survive, and the weights span 0.013 to 869. The normalization check comes out at 8.75 with a weighted mean of +0.29 K, so the weights cancel the tilt. Cloning on a forecast of the event outperforms cloning on the present state, as the Gate 3 persistence column predicted before these runs were made.

Slurm job 1180, 3 h 50 m · most extreme walker +7.33 K against an observed +7.72 K





Per-walker anomaly and probability mass, control run. The same panel as in each case section. No walker lies in the shaded band, so P is 0 and the statement the run supports is the deepest level it resolved, 2.6×10^{-4} . The 64 walkers hold 15 distinct masses and a weight ESS of 1.0, that is, one effective sample.